SUMMARY

This map book has been developed by CREASION Nepal as part of Output 4, Activity 9 of the Project CAP: Collaborative Approach for Preventing Plastic Leakages in Rivers of Nepal. The project is funded by PLEASE (Plastic-Free Rivers and Seas of South Asia) and implemented by the South Asia Cooperative Environment Program (SACEP), with support from the United Nations Office for Project Services (UNOPS) and the World Bank. Project CAP has developed a collaborative model to engage key stakeholders in establishing a robust plastic waste value chain.

The data presented in this map book was gathered through a multi-method approach, including Focus Group Discussions (FGDs), Key Informant Interviews (KIIs), GPS tracking for hotspot identification, and in-situ sampling to validate plastic waste generation. Additionally, plastic waste sampling facilitated an accurate analysis of solid waste composition found in municipal landfills. The primary objective of this map book is to support policymakers in identifying major plastic waste hotspots, understanding the types of plastic present, and pinpointing key leakage sources. Developed in close collaboration with each municipalities, the information presented ensures accuracy and relevance to local needs. By offering detailed insights into plastic waste composition, this map book serves as a valuable tool for municipalities to design effective waste management strategies and develop policy frameworks aimed at preventing plastic leakage into rivers.

HOW TO UNDERSTAND THIS MAP BOOK?

This map book is organized into two thematic sections: Administrative Maps and Plastic Waste Composition Maps. It visually presents plastic waste hotspots within the municipality, detailing plastic types, leakage sources, and distribution patterns. Additionally, it features visual descriptions of various plastic types to improve understanding and support effective waste management strategies.

Each title of this map reflects its respective theme. A legend is included in every map, explaining the symbols and icons used to represent various features. Additionally, each main map is accompanied by an inset map at the district level, highlighting the specific municipality or rural municipality featured.

The thematic administrative map displays major physical infrastructure alongside administrative boundaries sourced from the Survey Department of Nepal and OpenStreetMap (OSM). It also includes municipal demographic details such as total population, population density, total area, male-to-female population ratio, and total households. The plastic waste composition map incorporates Environmental Systems Research Institute (ESRI)'s topographic base map and land cover data from International Centre for Integrated Mountain Development (ICIMOD), along with solid waste composition data obtained through in-situ sampling at identified hotspots. To images from the hotspot locations.

METHODOLOGY OF THE MAP BOOK

This Map Book has been developed using field data collected through key informant interviews and focused group discussions. Primary data collection followed the "Guidelines for the Monitoring and Assessment of Plastic Litter in the Ocean" by the Joint Group of Experts on the Scientific Aspects of Marine Environmental Protection (GESAMP), published by UNEP in 2019, to ensure standardized methodologies. Additionally, a review of various literature sources provided secondary data on municipal waste. On-site sampling and site verification were conducted at the waste hotspots to further clarify waste management scenarios.

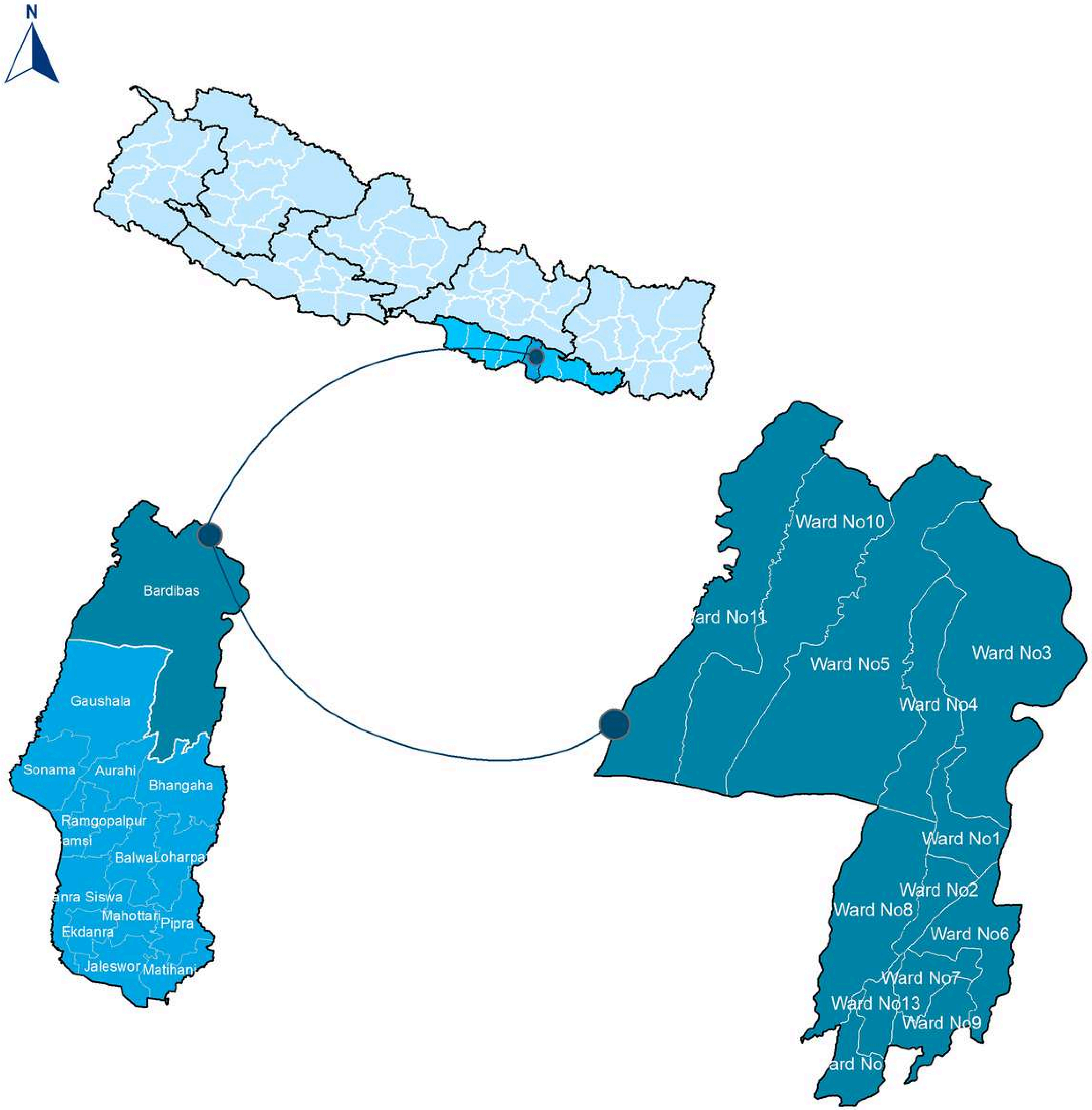
CONSENT FOR THIS MAP BOOK

This Map Book has been published in coordination with the respective municipalities, with the full consent of the respective stakeholders and local authorities involved in waste data collection and field verification. It is a collective effort of individuals engaged in gathering and verifying waste data at the ground level. This Map Book ensures the agreement and approval of all stakeholders, both directly and indirectly associated.

LIMITATION OF THIS MAP BOOK

The maps in the map book have been developed using primary data collected through field surveys, Focus Group Discussions (FGDs), and Key Informant Interviews (KIIs) conducted during the sampling process. Additionally, secondary data sources, including the Department of Survey Nepal, OpenStreetMap, ESRI, municipal databases, and the Forest Research and Training Centre, have been incorporated.

TABLE OF CONTENT

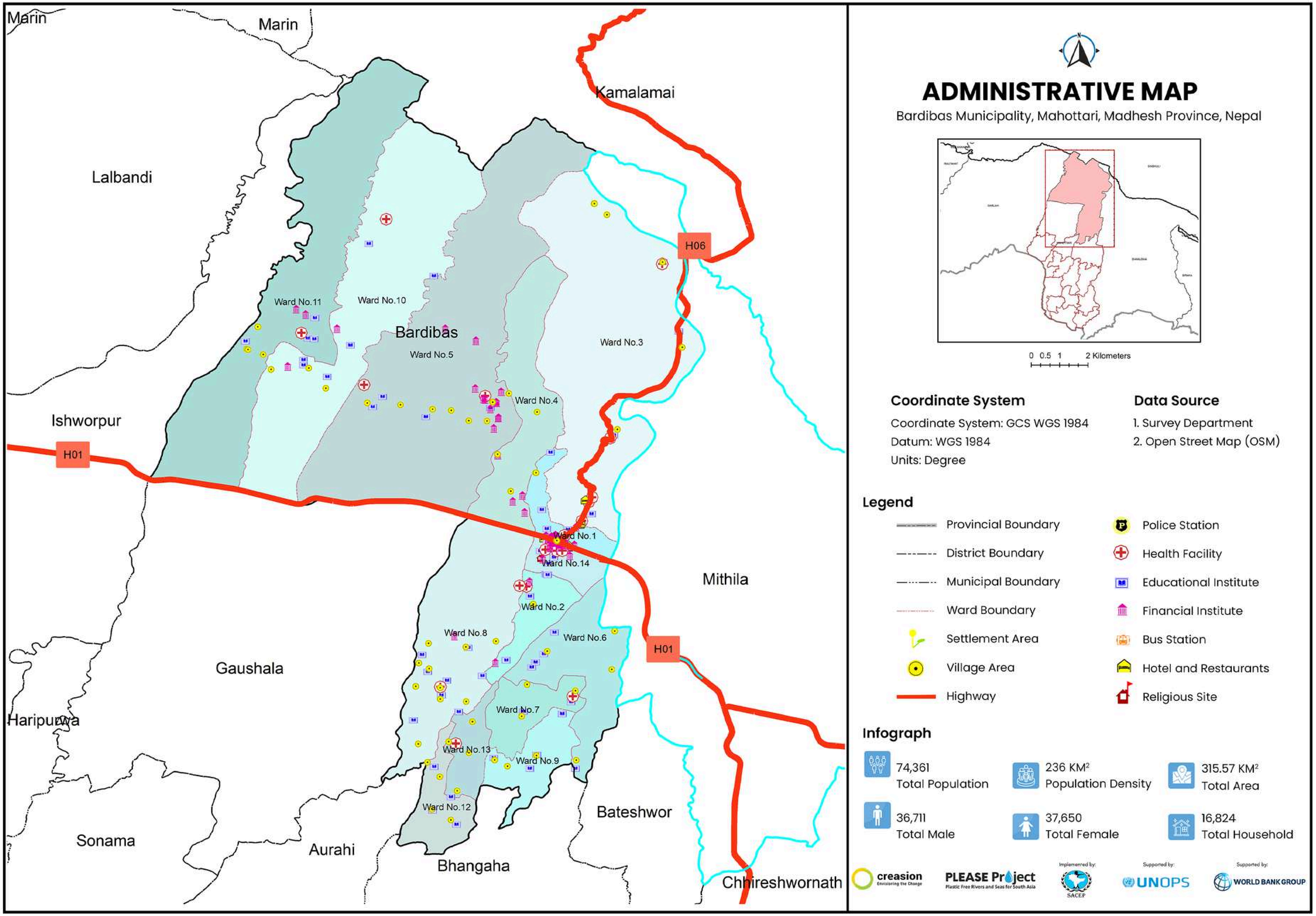


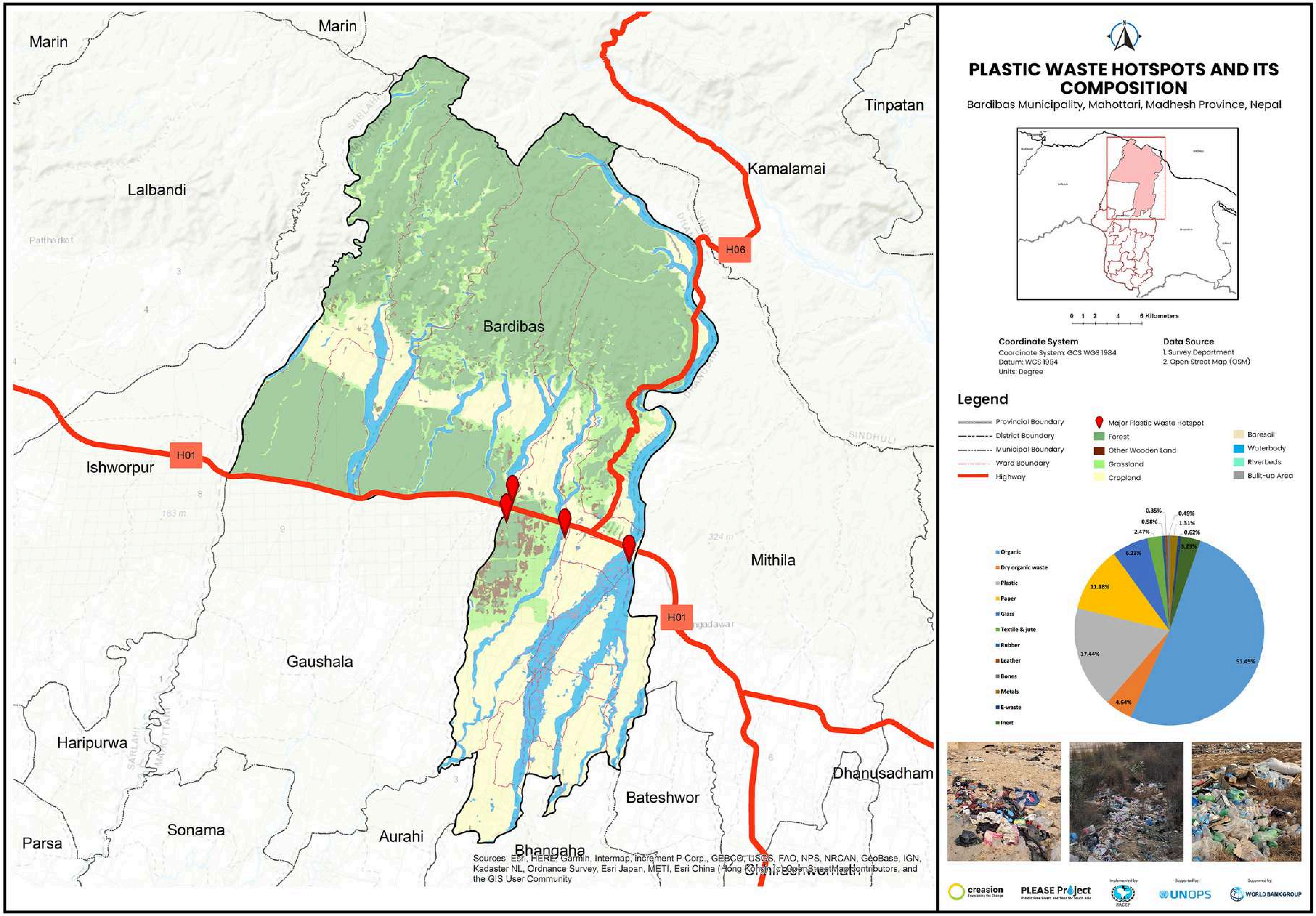
INTRODUCTION

Bardibas is a vibrant municipality located in the Mahottari District of Nepal, nestled at the foothills of the Chure range. Strategically positioned along the East-West Highway, it serves as a vital link between the Kathmandu Valley, Madhesh Province, and Koshi Province. Covering a total area of 315.57 square kilometers, Bardibas is divided into 14 wards and has a population of 68,353, with 16,824 households and the literacy rate of 67%. (Census, 2021).

Located at an altitude of approximately 80 meters (262 feet) above sea level, Bardibas sits at coordinates 26.98°N, 85.90°E. The municipality is enriched by one major river—Ratu—that lies East to Bardibas. The river play a crucial role in supporting local agriculture, providing essential water resources for irrigation, and sustaining the livelihoods of the community. The fertile land and abundant water resources make Bardibas a key agricultural hub in the region, where farming is central to the local economy and way of life.

The river not only supports agriculture but also contributes to the region's ecosystem. However, like many rivers in Nepal, they are increasingly threatened by plastic waste leakage, which poses a serious environmental challenge. Protecting these rivers from pollution is essential for maintaining the health of the community and the natural resources that sustain it.


ADMINISTRATIVE MAP
Bardibas Municipality, Mahottari, Madhesh Province, Nepal
0 0.5 1 2 Kilometers
Coordinate System
Coordinate System: GCS WGS 1984
Datum: WGS 1984
Units: Degree
Data Source
1. Survey Department
2. Open Street Map (OSM)
Legend
Provincial Boundary
District Boundary
Municipal Boundary
Ward Boundary
Settlement Area
Village Area
Highway
Police Station
Health Facility
Educational Institute
Financial Institute
Bus Station
Hotel and Restaurants
Religious Site
Infograph
74,361 Total Population
236 KM² Population Density
315.57 KM² Total Area
36,711 Total Male
37,650 Total Female
16,824 Total Household
creasion
PLEASE Project
Implemented by:
SACEP
Supported by:
UNOPS
Supported by:
WORLD BANK GROUP
Marin
Marin
Kamalamai
Lalbandi
Ishworpur
H01
H06
H01
Bardibas
Ward No.11
Ward No.10
Ward No.5
Ward No.3
Ward No.4
Ward No.1
Ward No.14
Ward No.2
Ward No.8
Ward No.6
Ward No.7
Ward No.13
Ward No.9
Ward No.12
Mithila
Gaushala
Haripurwa
Sonama
Aurahi
Bhangaha
Bateshwor
Chhireshwornath



PLASTIC WASTE HOTSPOTS AND ITS COMPOSITION
Bardibas Municipality, Mahottari, Madhesh Province, Nepal
0 1 2 4 6 Kilometers
Coordinate System
Coordinate System: GCS WGS 1984
Datum: WGS 1984
Units: Degree
Data Source
1. Survey Department
2. Open Street Map (OSM)
Legend
Provincial Boundary
District Boundary
Municipal Boundary
Ward Boundary
Highway
Major Plastic Waste Hotspot
Forest
Other Wooden Land
Grassland
Cropland
Baresoil
Waterbody
Riverbeds
Built-up Area
Organic
Dry organic waste
Plastic
Paper
Glass
Textile & jute
Rubber
Leather
Bones
Metals
E-waste
Inert
51.45%
4.64%
17.44%
11.18%
6.23%
2.47%
0.58%
0.35%
0.49%
1.31%
0.62%
3.23%
Marin
Marin
Tinpatan
Kamalamai
Lalbandi
Patharkot
H06
Bardibas
H01
Ishworpur
Mithila
H01
Gaushala
Haripurwa
Dhanusadham
Bateshwor
Sonama
Parsa
Aurahi
Bhangaha
Sources: Esri, HERE, Garmin, Intermap, increment P Corp., GEBCO, USGS, FAO, NPS, NRCAN, GeoBase, IGN, Kadaster NL, Ordnance Survey, Esri Japan, METI, Esri China (Hong Kong), (c) OpenStreetMap contributors, and the GIS User Community
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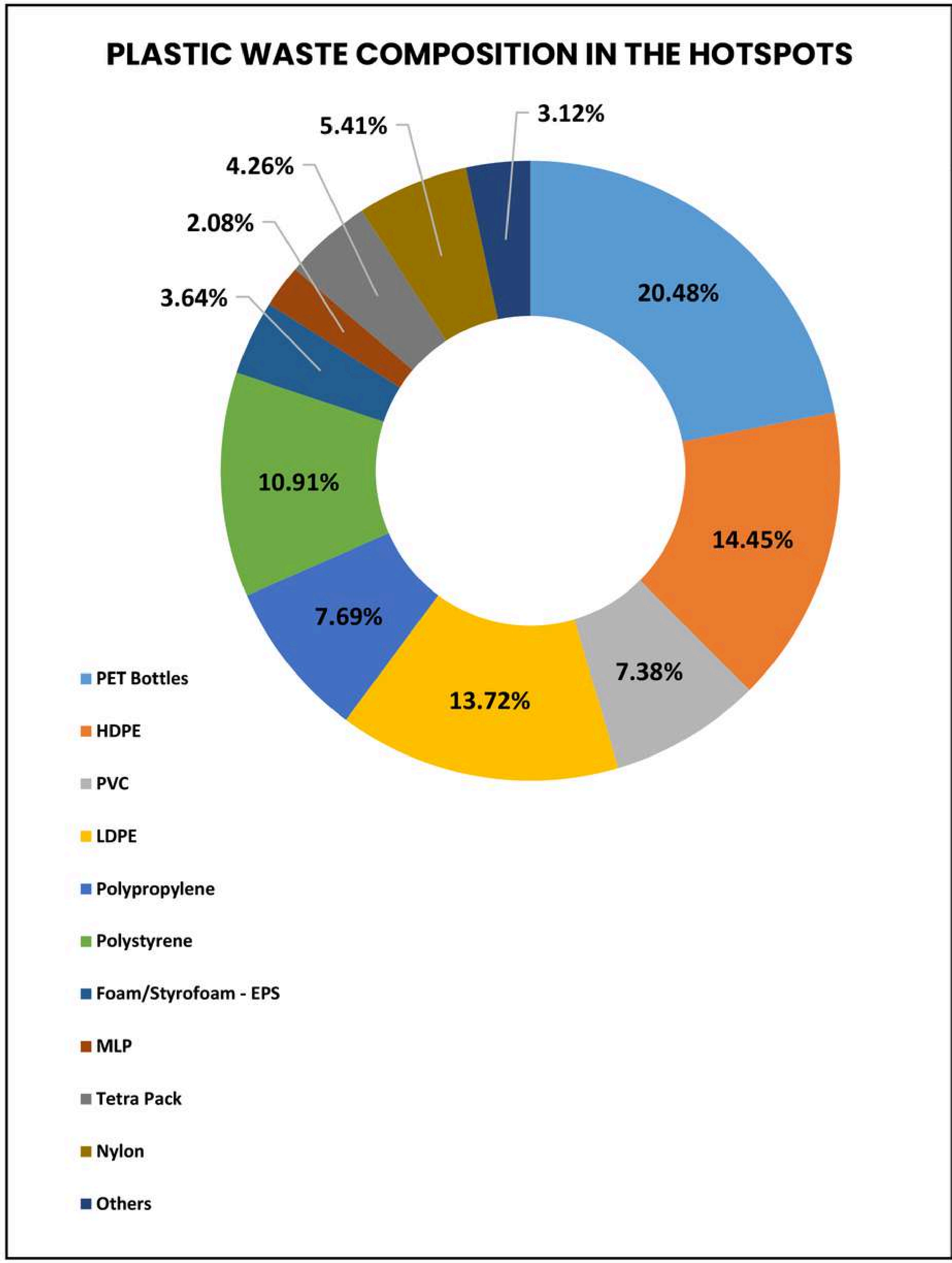
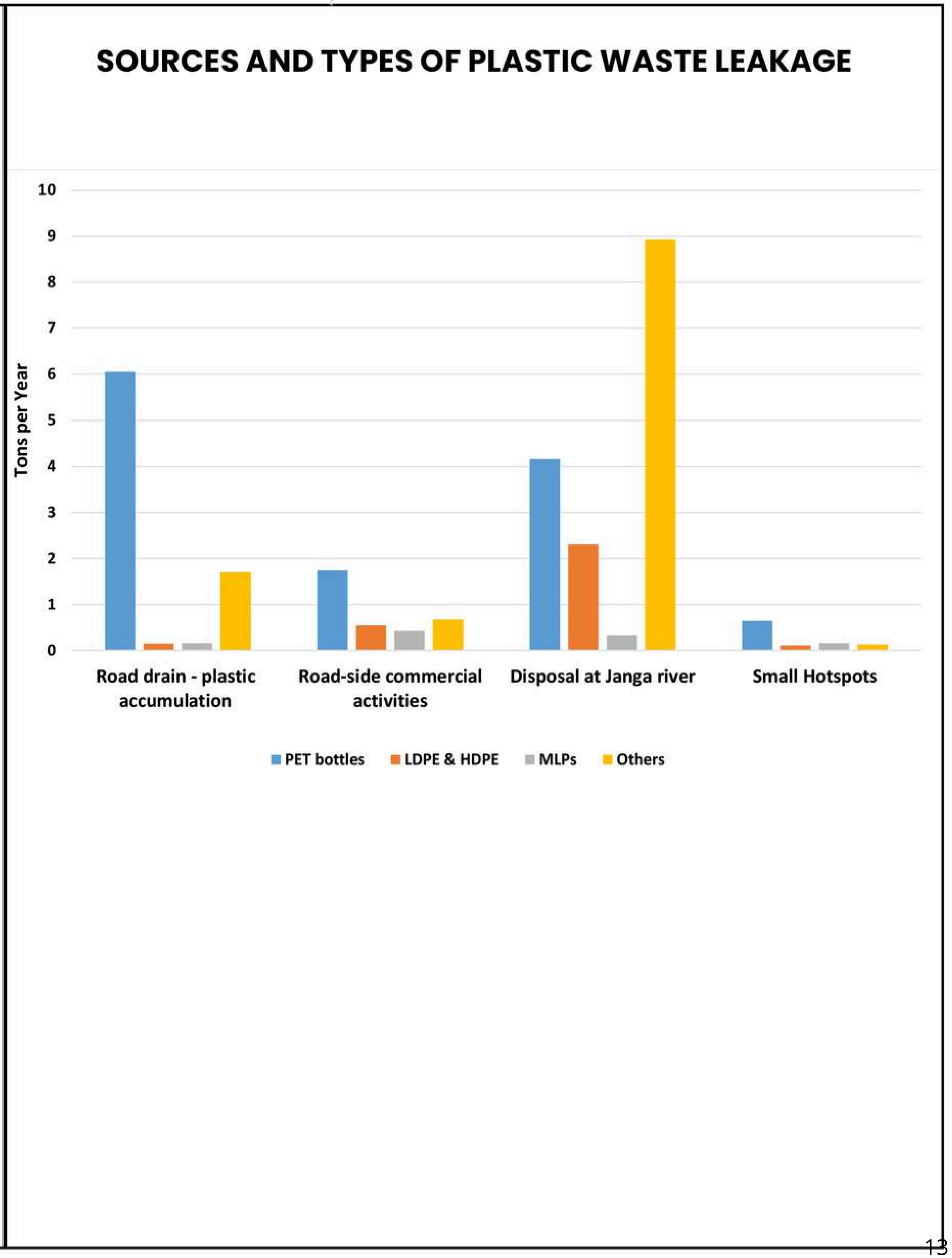
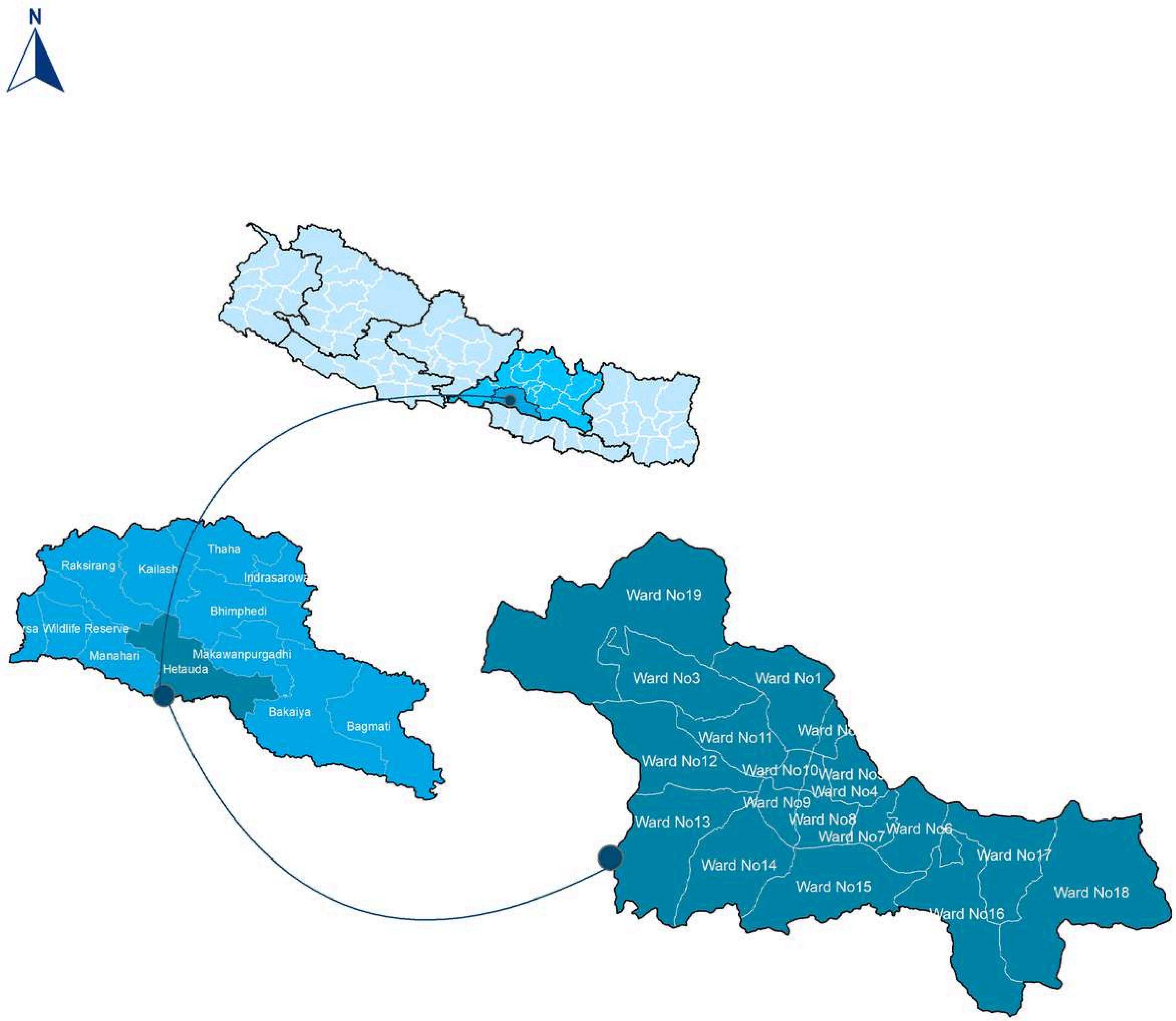
PLEASE Project
UNOPS
WORLD BANK GROUP


PLASTIC WASTE COMPOSITION IN THE HOTSPOTS
5.41%
3.12%
4.26%
2.08%
3.64%
20.48%
10.91%
14.45%
7.69%
7.38%
13.72%
PET Bottles
HDPE
PVC
LDPE
Polypropylene
Polystyrene
Foam/Styrofoam - EPS
MLP
Tetra Pack
Nylon
Others


SOURCES AND TYPES OF PLASTIC WASTE LEAKAGE
Tons per Year
10
9
8
7
6
5
4
3
2
1
0
Road drain - plastic accumulation
Road-side commercial activities
Disposal at Janga river
Small Hotspots
PET bottles
LDPE & HDPE
MLPs
Others

13



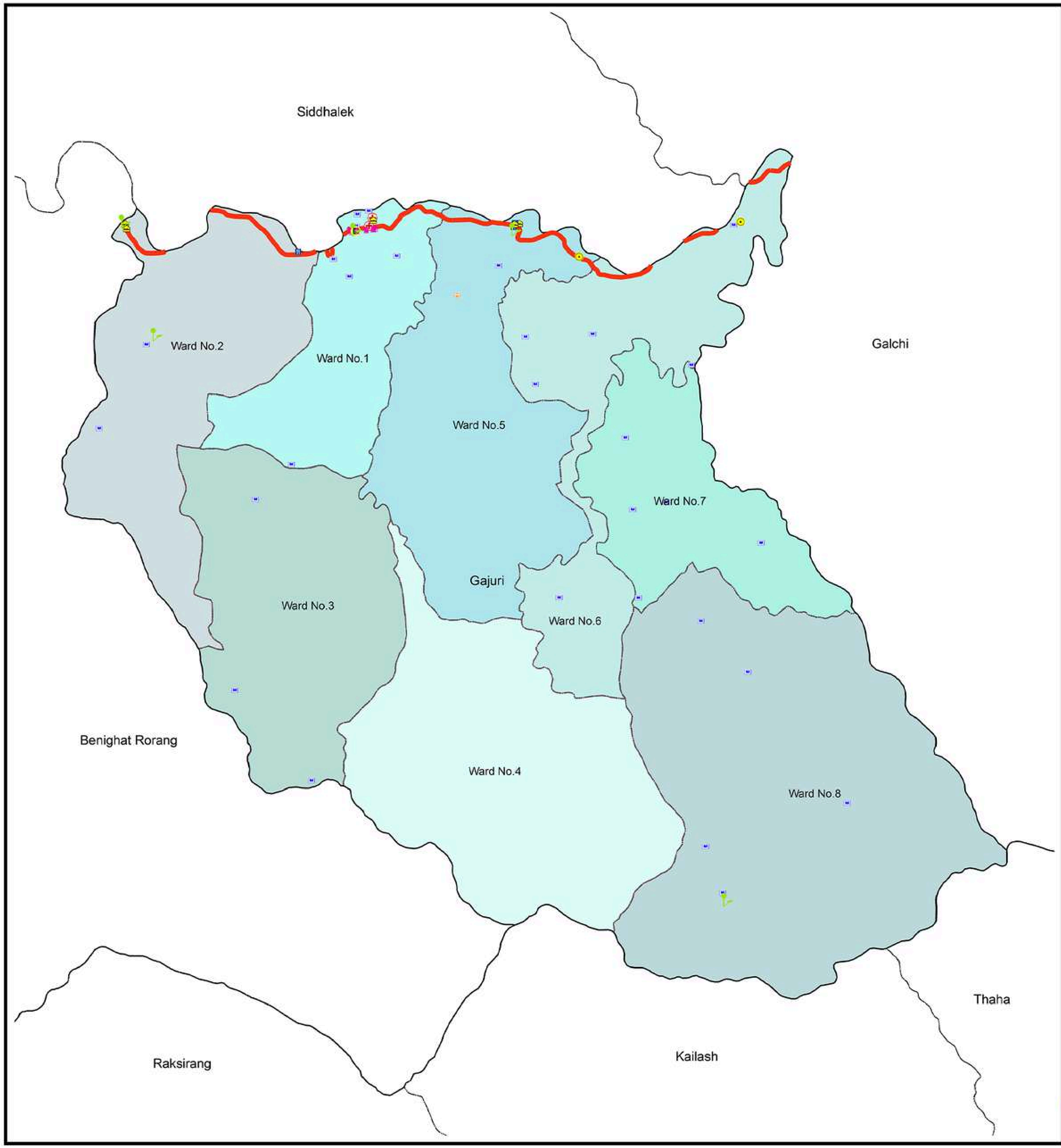
INTRODUCTION

Gajuri Rural Municipality is located in the North-Western part of Dhading district, nestled to the south of the Trishuli River at coordinates 27°46'N 84°51'E, with an average altitude of 574 meters above sea level. Gajuri with the area of 139 square kilometers, is divided into 8 wards, each scattered across this expansive landscape. According to 2021 census, the municipality is home to a population of 28,634 people, residing in 6,839 households, with a literacy rate of 72.9%.

The climate of Gajuri is generally temperate, characterized by cool winters, warm summers, and significant rainfall during the monsoon season. These conditions contribute to the fertile lands of the region, which support the cultivation of key crops such as maize, rice, and various vegetables. Agriculture remains a central part of Gajuri's economy and way of life.

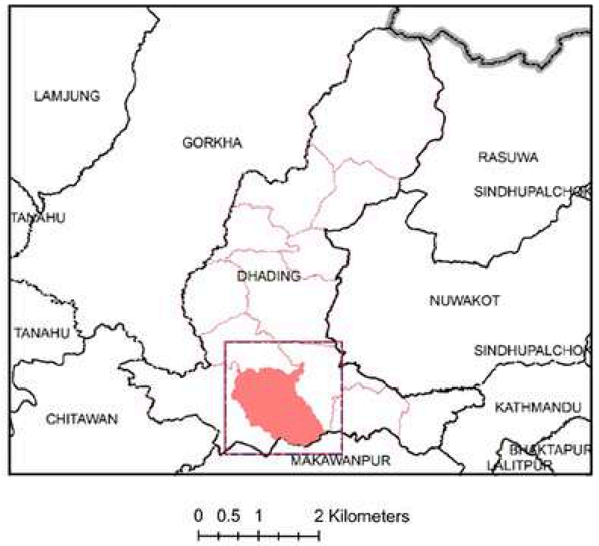
One major river, the Trishuli River, flow through the municipality, adding to the region's natural beauty and ecological significance. The Trishuli River, which runs through the western part of Gajuri, holds deep cultural, economic, and environmental importance. Similarly, the Roshi River, situated along the eastern side, plays a vital role in sustaining local agriculture by providing irrigation and supporting aquatic ecosystems. These rivers are essential to the community, both for their ecological functions and as sources of water for agriculture, consumption, and daily life.

Together, these rivers and the fertile land forms the foundation of Gajuri's agriculture-based economy and contribute to the vibrant local culture and lifestyle.



ADMINISTRATIVE MAP

Gajuri Rural Municipality, Dhading, Bagmati Province, Nepal



Coordinate System

Coordinate System: GCS WGS 1984
Datum: WGS 1984
Units: Degree

Data Source

1. Survey Department
2. Open Street Map (OSM)

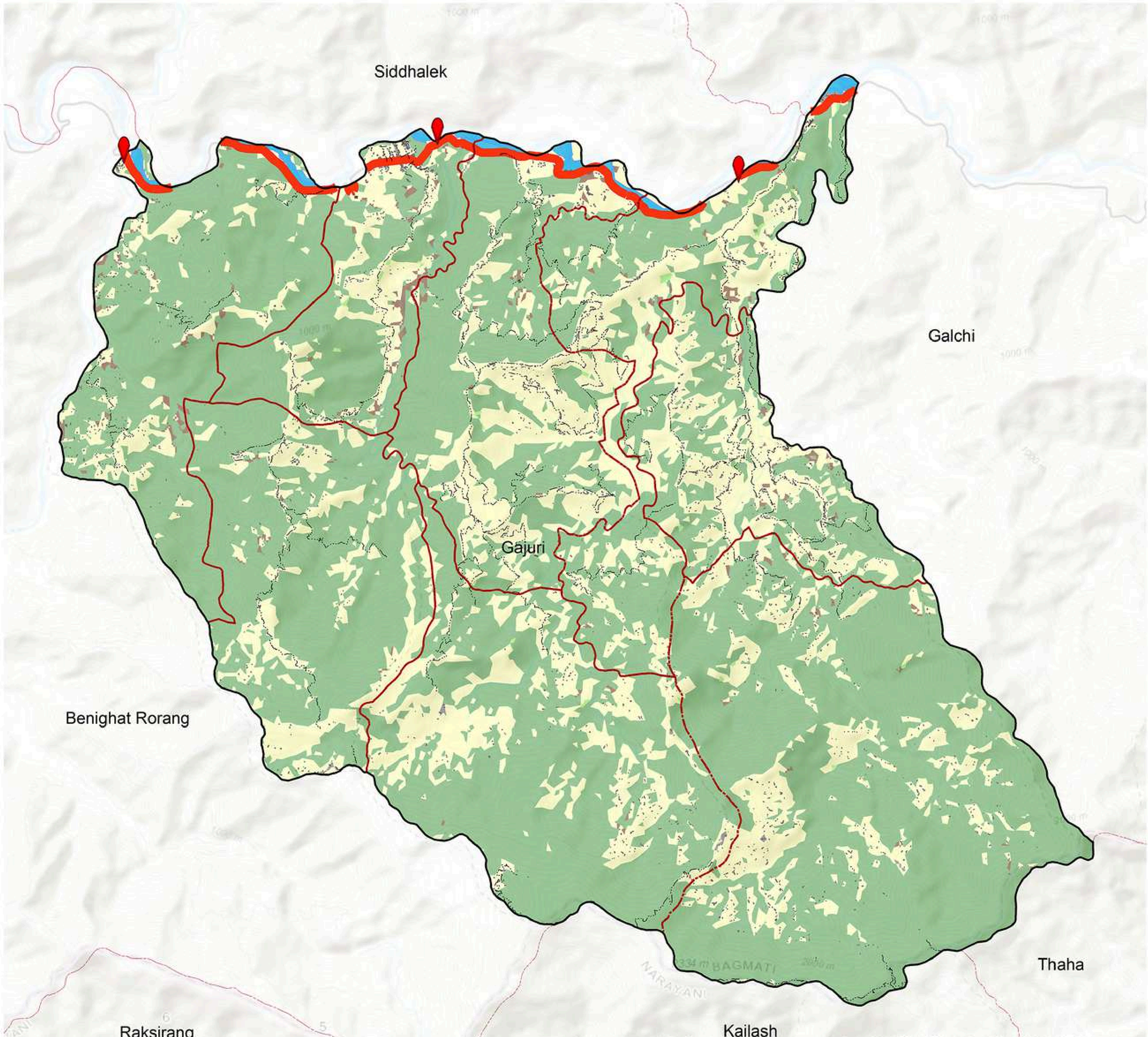
Legend

- Provincial Boundary
- District Boundary
- Municipal Boundary
- Ward Boundary
- Settlement Area
- Village Area
- Highway
- Police Station
- Health Facility
- Educational Institute
- Financial Institute
- Bus Station
- Hotel and Restaurants
- Religious Site

Infograph

- 28,634 Total Population
- 207 KM² Population Density
- 139 KM² Total Area
- 14,326 Total Male
- 14,308 Total Female
- 6,839 Total Household

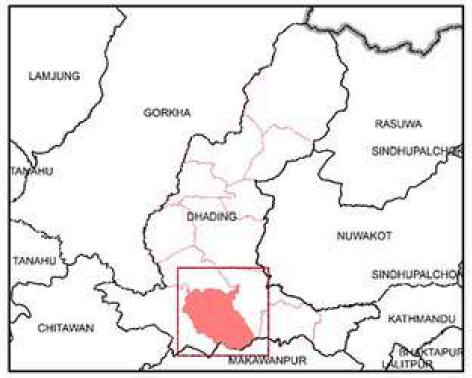
Implemented by: Supported by: Supported by:



Sources: Esri, HERE, Garmin, Intermap, increment P Corp., GEBCO, USGS, FAO, NPS, NRCAN, GeoBase, IGN, Kadaster NL, Ordnance Survey, Esri Japan, METI, Esri China (Hong Kong), (c) OpenStreetMap contributors, and the GIS User Community

PLASTIC WASTE HOTSPOTS AND ITS COMPOSITION

Gajuri Rural Municipality, Dhading, Bagmati Province, Nepal



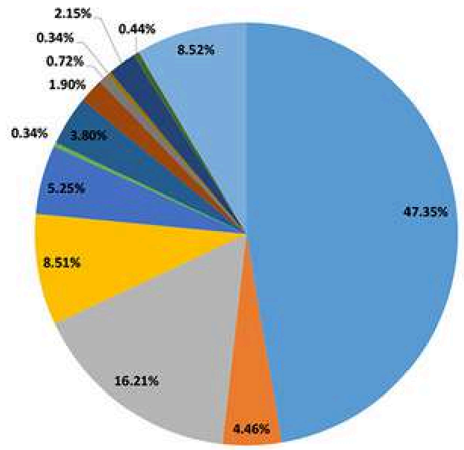
0 1 2 4 6 Kilometers

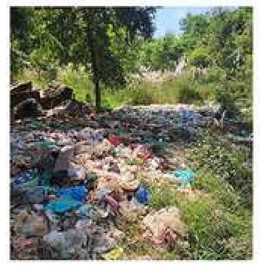
Coordinate System
Coordinate System: GCS WGS 1984
Datum: WGS 1984
Units: Degree

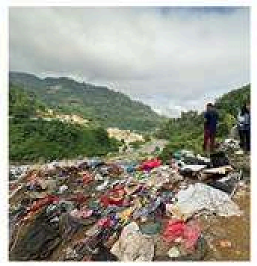
Data Source
1. Survey Department
2. Open Street Map (OSM)

Legend

- Provincial Boundary
- District Boundary
- Municipal Boundary
- Ward Boundary
- Highway
- Major Plastic Waste Hotspot
- Forest
- Other Wooden Land
- Grassland
- Cropland
- Baresoil
- Waterbody
- Riverbeds
- Built-up Area







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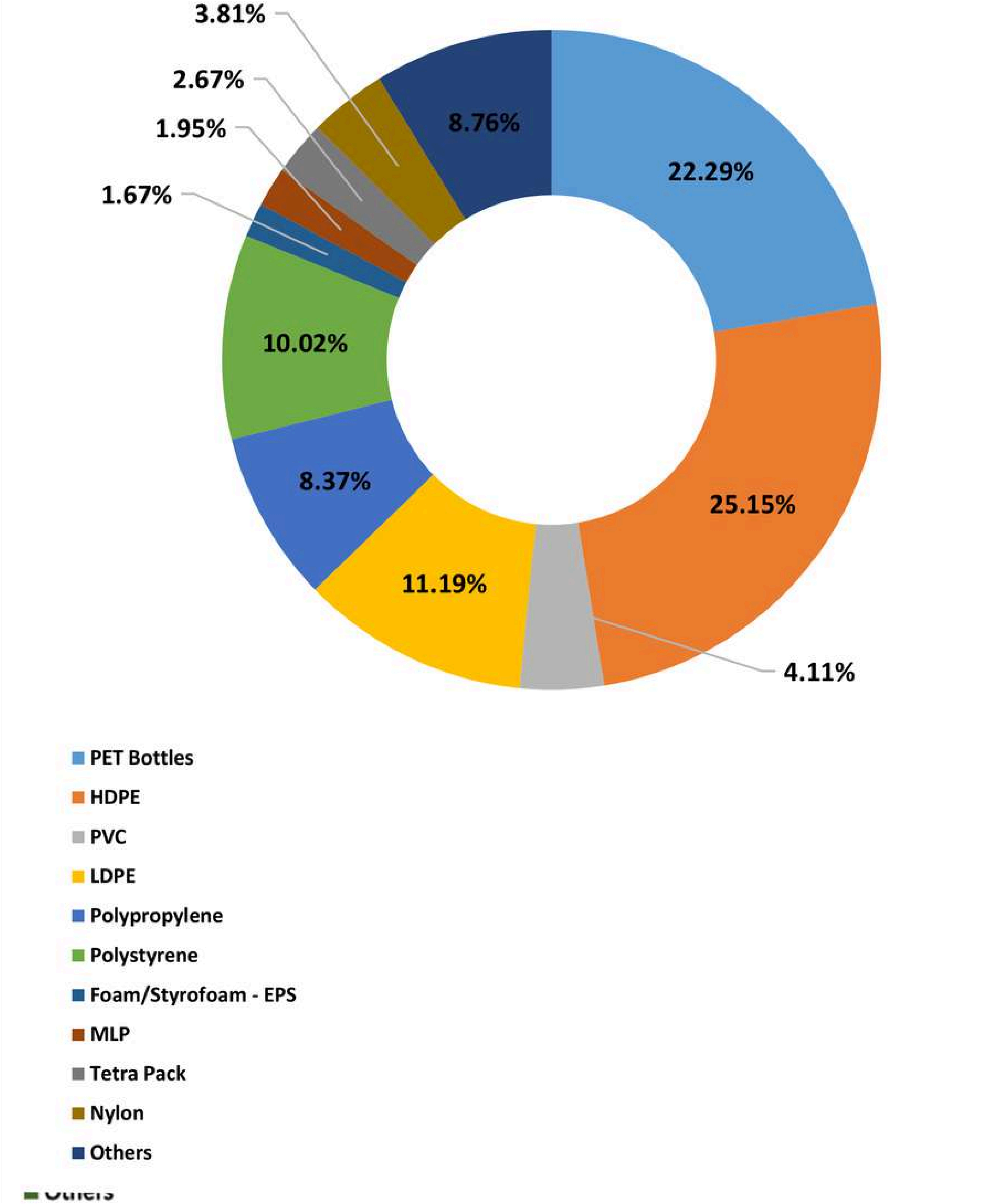
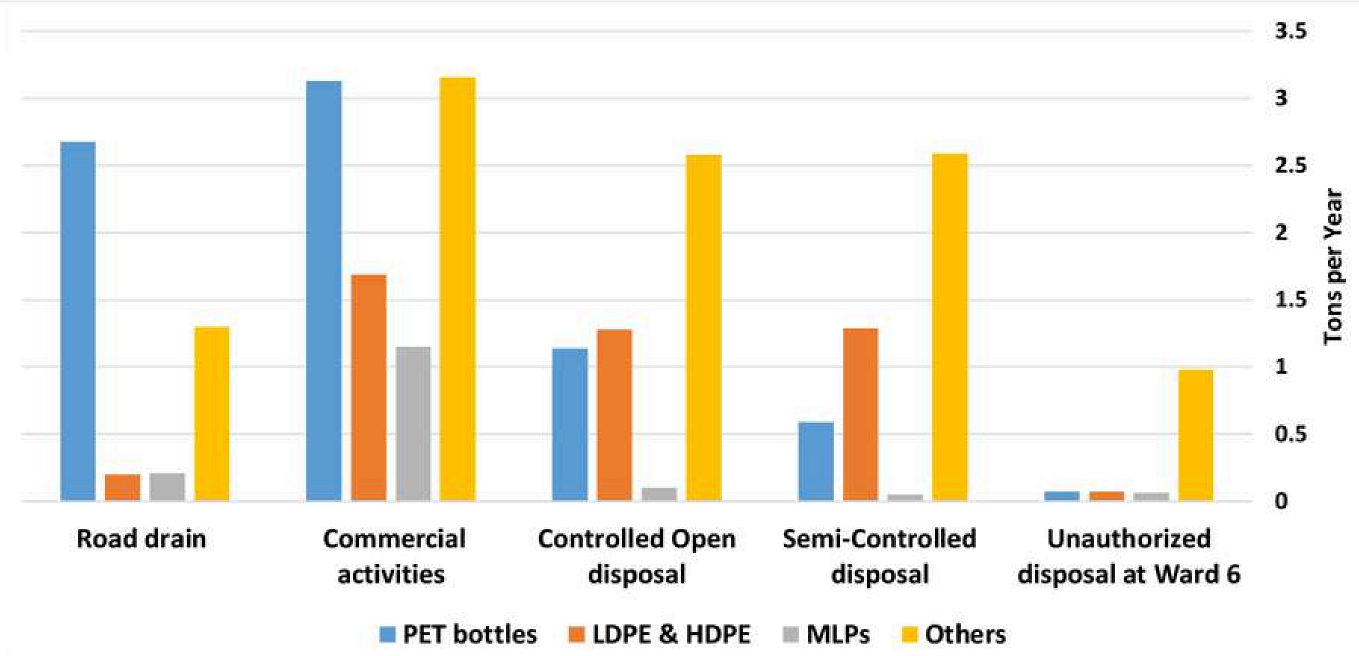
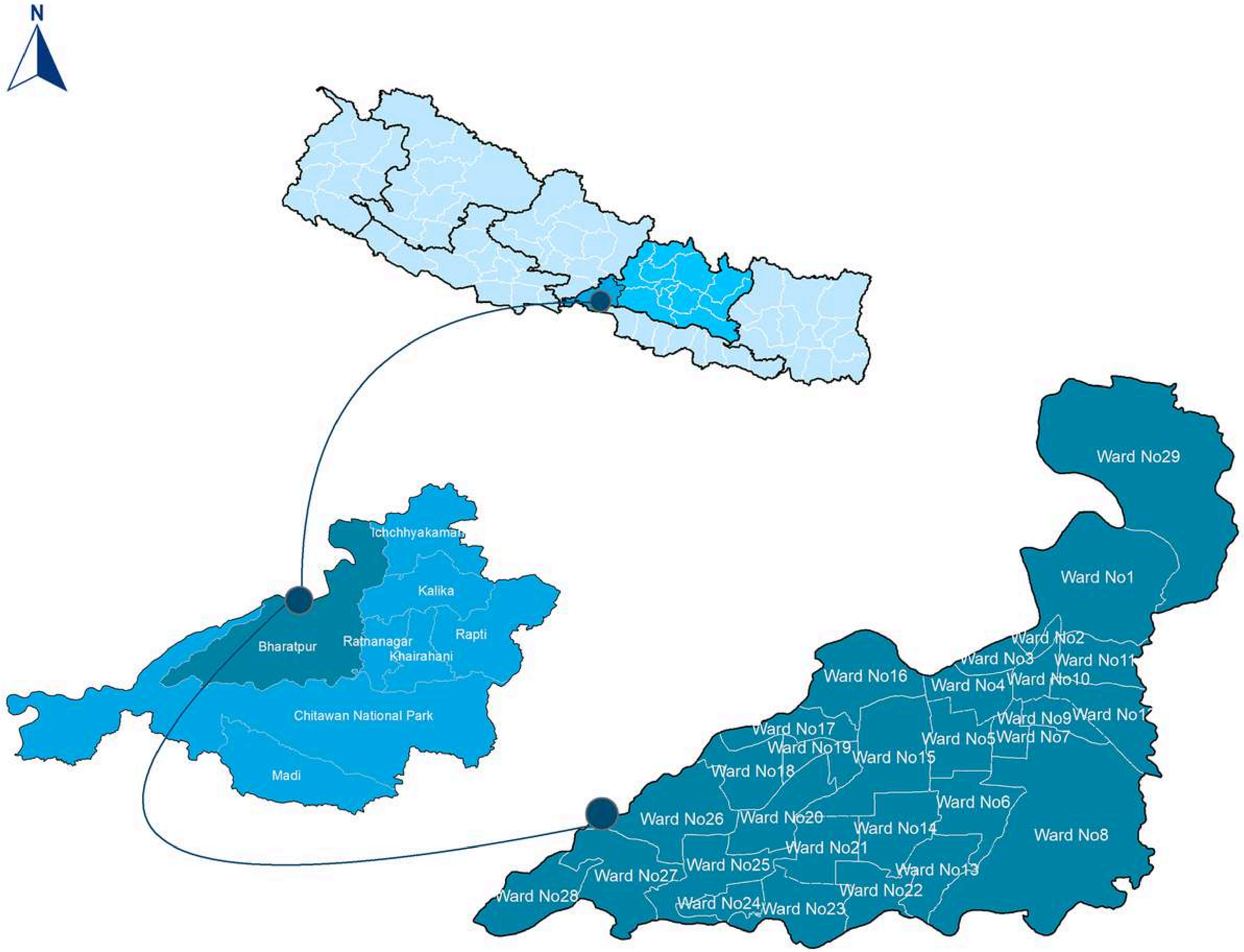
PLASTIC WASTE COMPOSITION IN THE HOTSPOTS



SOURCES AND TYPES OF PLASTIC WASTE LEAKAGE



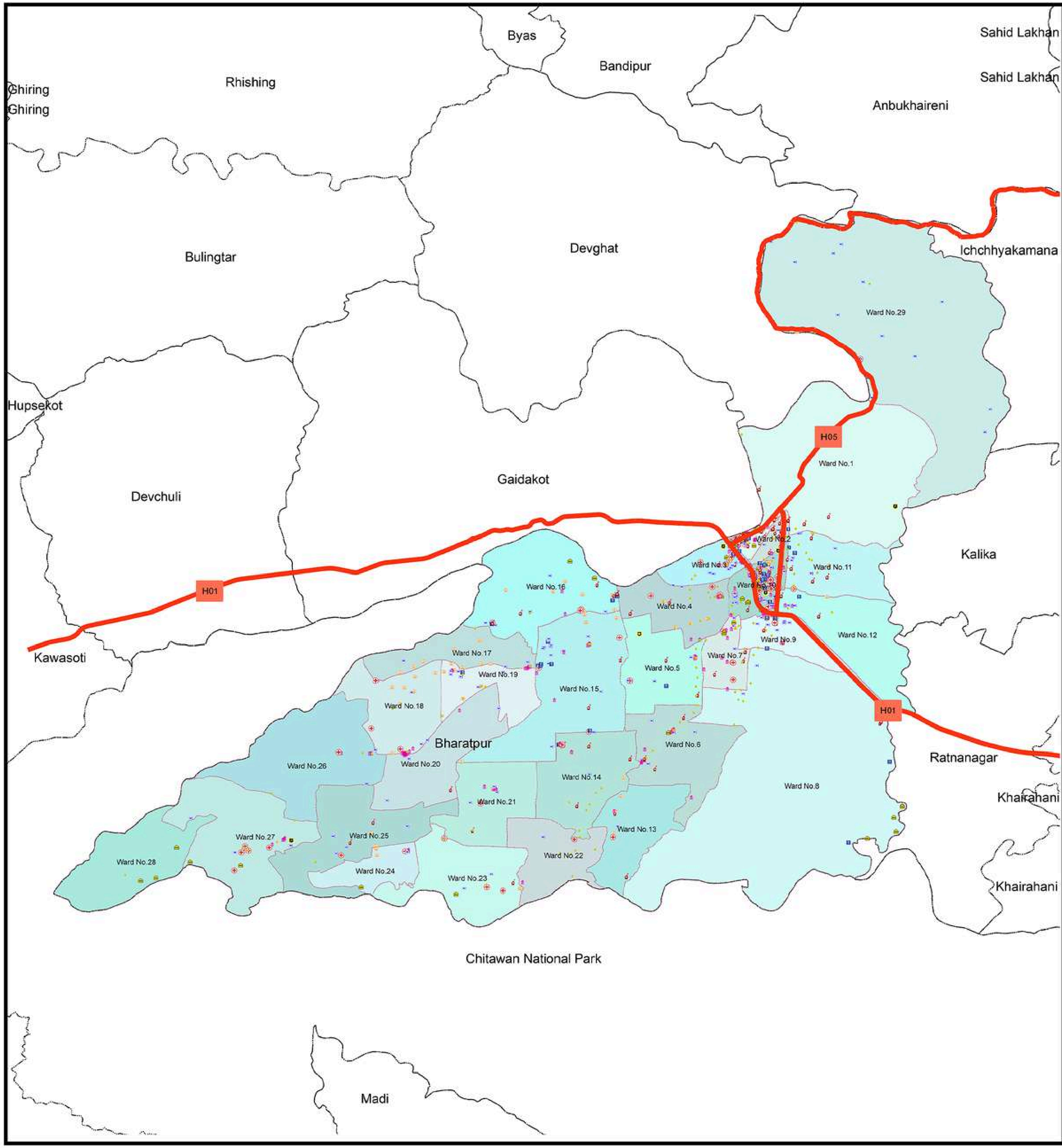


INTRODUCTION

Bharatpur Metropolitan City, located in the central-southern part of Nepal, serves as the administrative center of Chitwan District. As the fifth largest city in the country, it is a vibrant commercial hub with a population of 369,268 and 96,591 households (Census, 2021). Bharatpur's strategic location on the left bank of the Narayani River, along key transport routes like the Mahendra Highway and the Kathmandu-Birgunj road corridor, makes it an essential transit point for trade and travel. Spanning an area of 432.95 square kilometers, the city has a literacy rate of 87.2%.

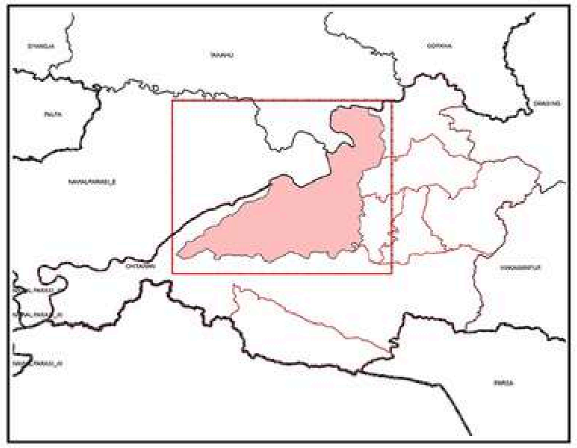
Nestled in Nepal's fertile Terai region, Bharatpur benefits from a warm climate and alluvial soil, making it ideal for agriculture. The region is vital to the country's food production, supporting the cultivation of rice, maize, wheat, and sugarcane. Bharatpur itself is situated at an altitude ranging from 140 to 390 meters above sea level, with the surrounding fertile plains playing a crucial role in the local economy and livelihood of its people.

The city is home to two major rivers: the Rapti and Narayani Rivers. The Rapti River, located near Bharatpur, plays a crucial role in irrigation, supporting local agriculture while also sustaining the ecosystem of the nearby Chitwan National Park. The Narayani River, one of Nepal's most significant waterways, provides irrigation and water supply to surrounding farms, serves as an important transportation route, and holds cultural and religious significance for local communities. These rivers are vital lifelines for Bharatpur and its surrounding areas, contributing to the region's agriculture, economy, and ecological balance.



ADMINISTRATIVE MAP

Bharatpur Metropolitan City, Chitwan, Bagmati Province, Nepal



0 0.5 1 2 Kilometers

Coordinate System

Coordinate System: GCS WGS 1984
Datum: WGS 1984
Units: Degree

Data Source

1. Survey Department
2. Open Street Map (OSM)

Legend

- Provincial Boundary
- District Boundary
- Municipal Boundary
- Ward Boundary
- Settlement Area
- Village Area
- Highway
- Police Station
- Health Facility
- Educational Institute
- Financial Institute
- Bus Station
- Hotel and Restaurants
- Religious Site

Infograph

369,268 Total Population
853 KM² Population Density
433 KM² Total Area
95,678 Total Male
190,371 Total Female
96,591 Total Household

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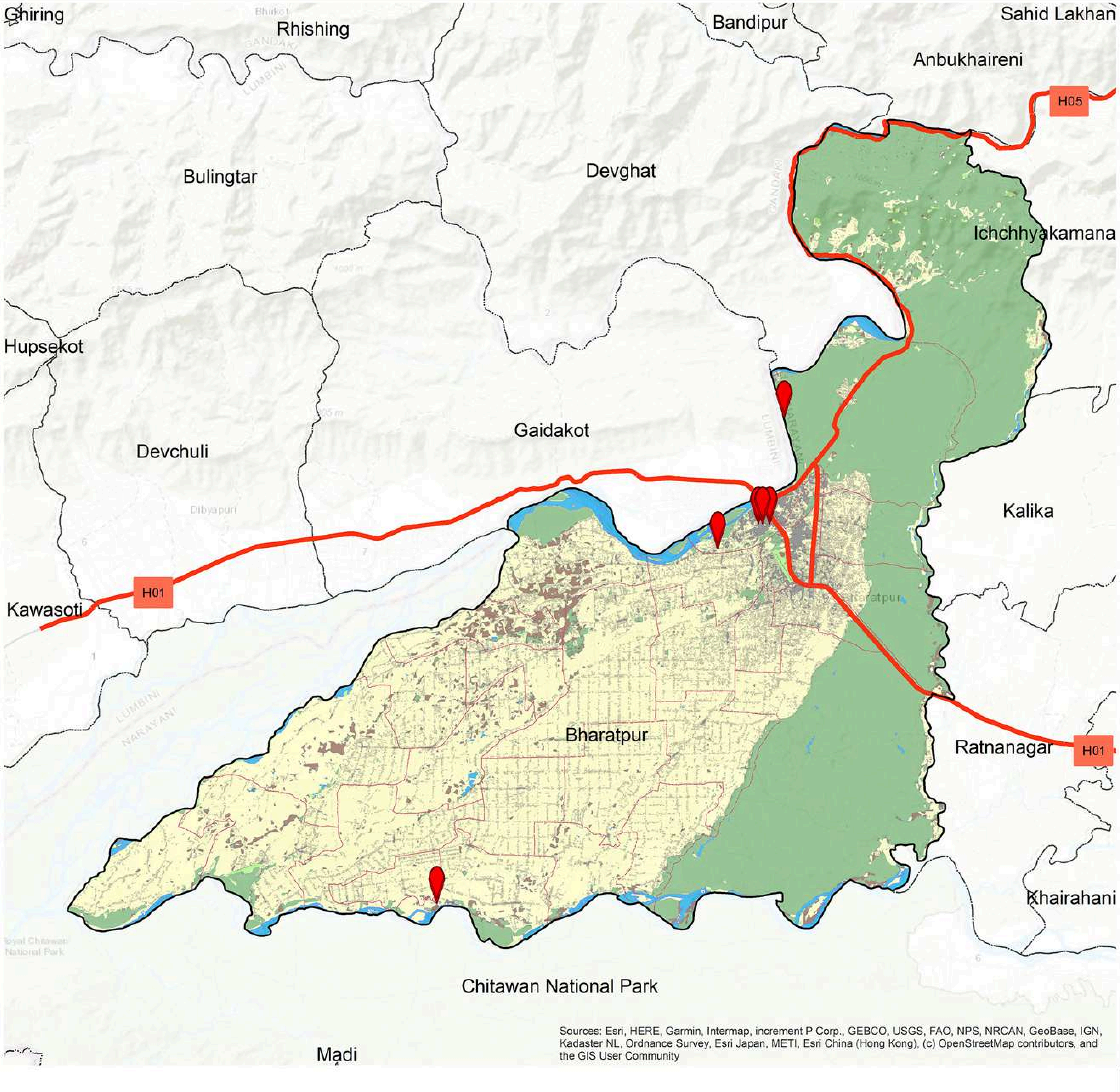
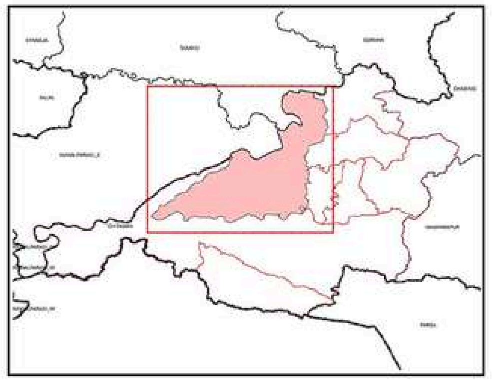
Supported by: WORLD BANK GROUP



PLASTIC WASTE HOTSPOTS AND ITS COMPOSITION

Bharatpur Metropolitan City, Chitwan, Bagmati Province, Nepal



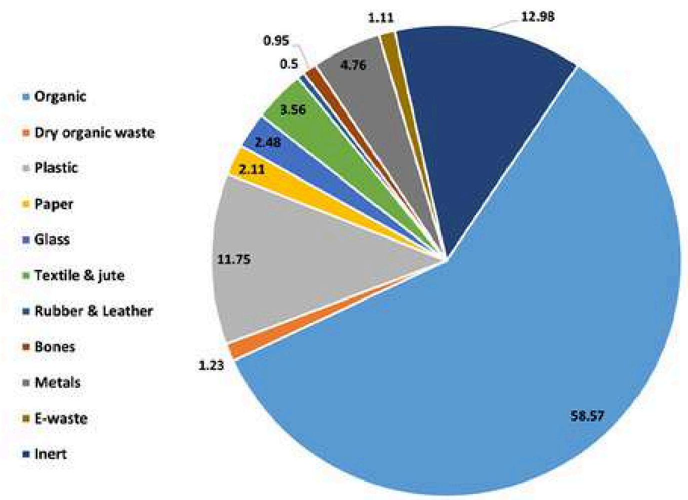
0 1 2 4 6 Kilometers

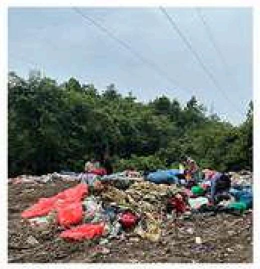
Coordinate System
Coordinate System: GCS WGS 1984
Datum: WGS 1984
Units: Degree

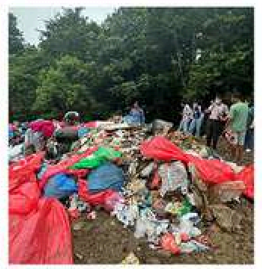
Data Source
1. Survey Department
2. Open Street Map (OSM)

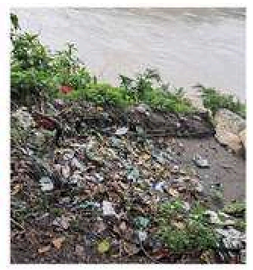
Legend

- Provincial Boundary
- District Boundary
- Municipal Boundary
- Ward Boundary
- Highway
- Major Plastic Waste Hotspot
- Forest
- Other Wooden Land
- Grassland
- Cropland
- Baresoil
- Waterbody
- Riverbeds
- Built-up Area









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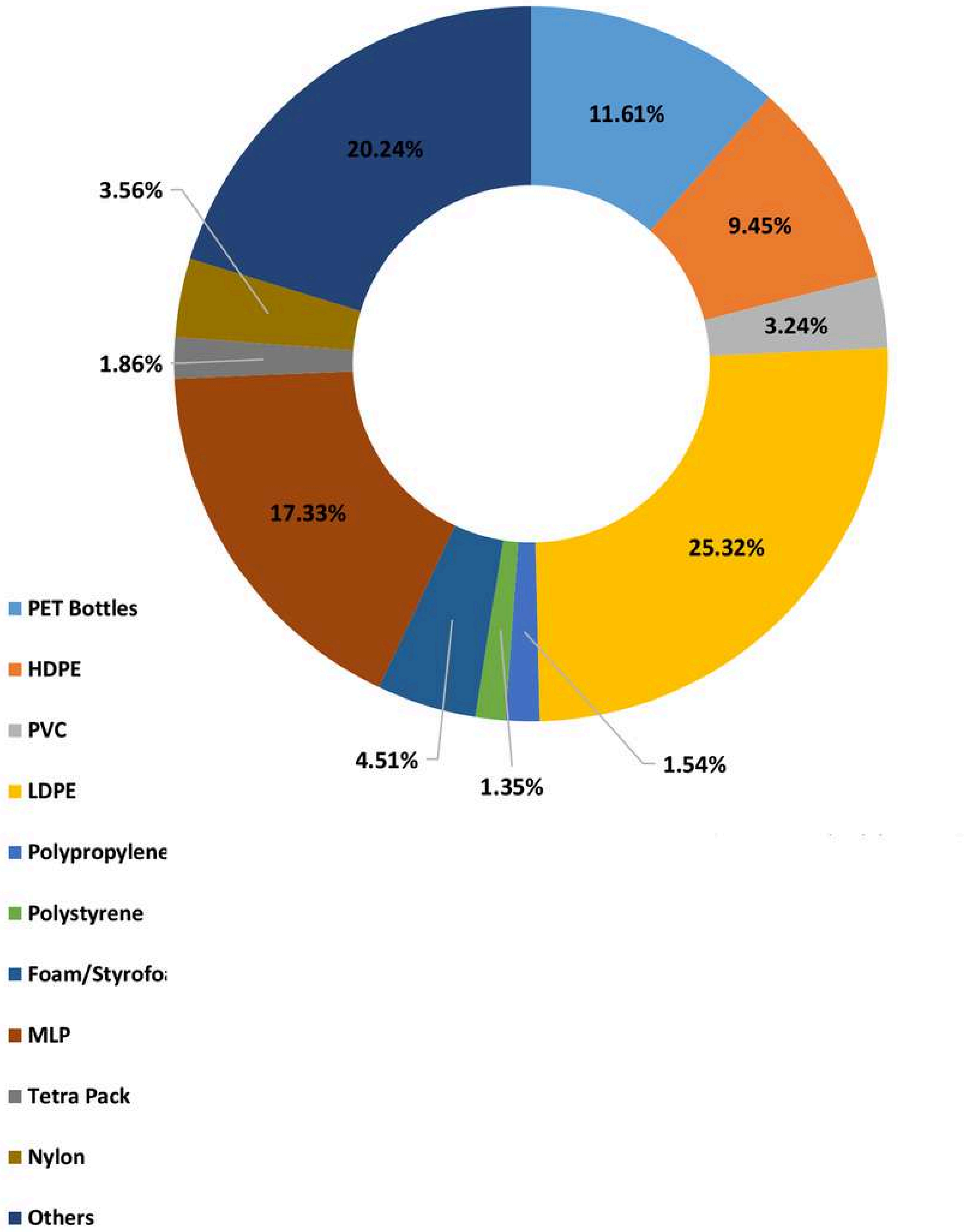
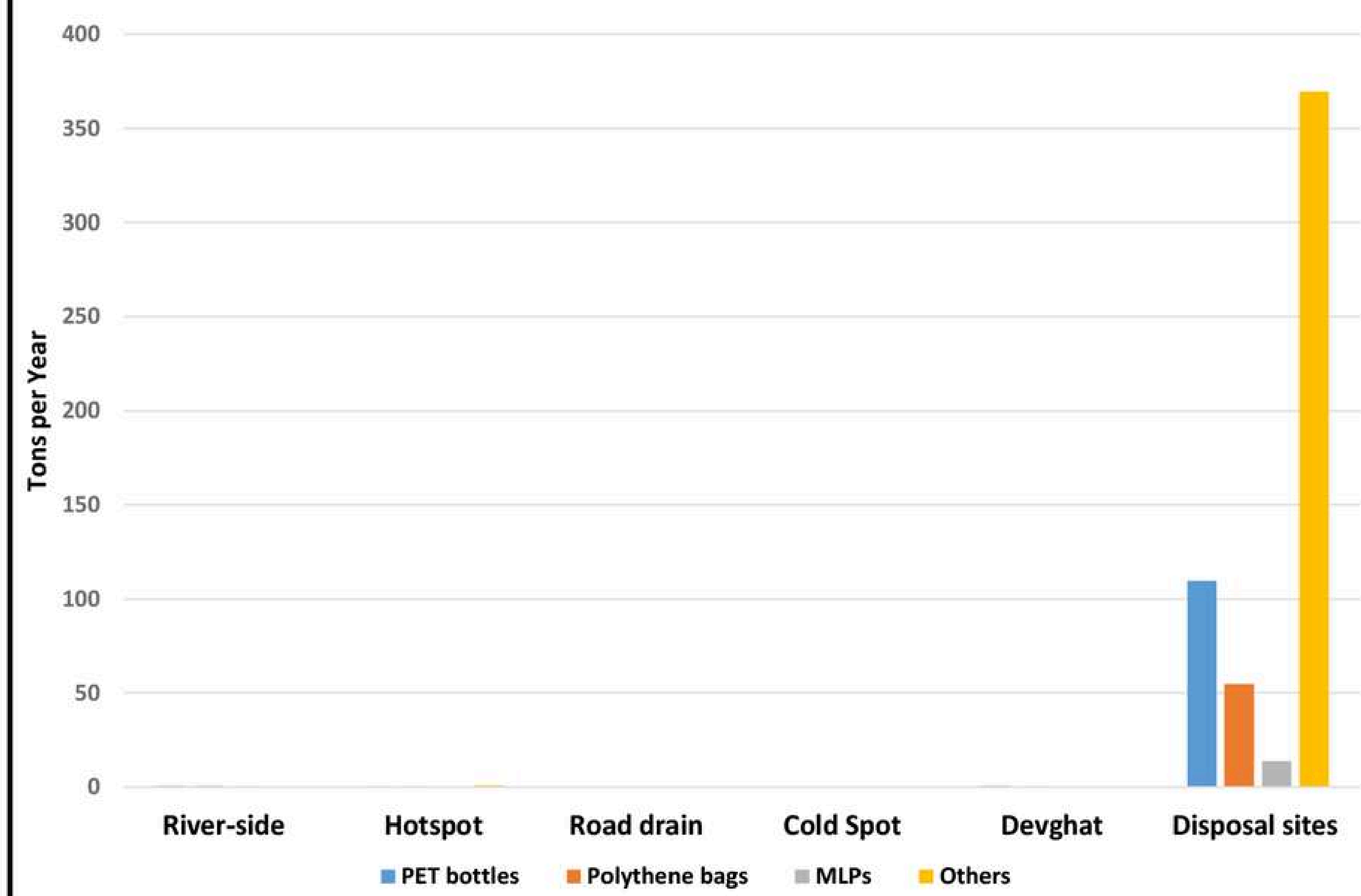
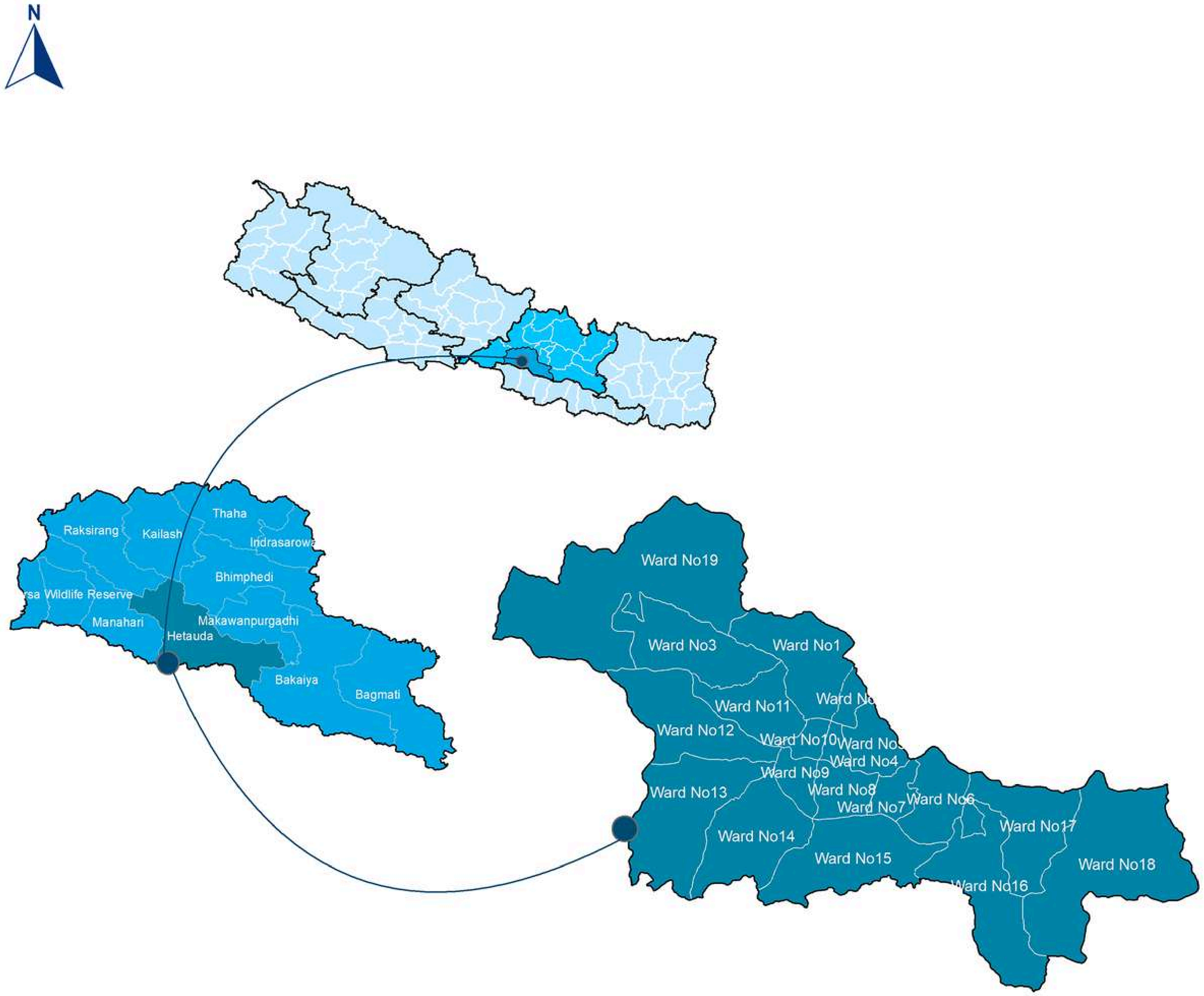
PLASTIC WASTE COMPOSITION IN THE HOTSPOTS



SOURCES AND TYPES OF PLASTIC WASTE LEAKAGE



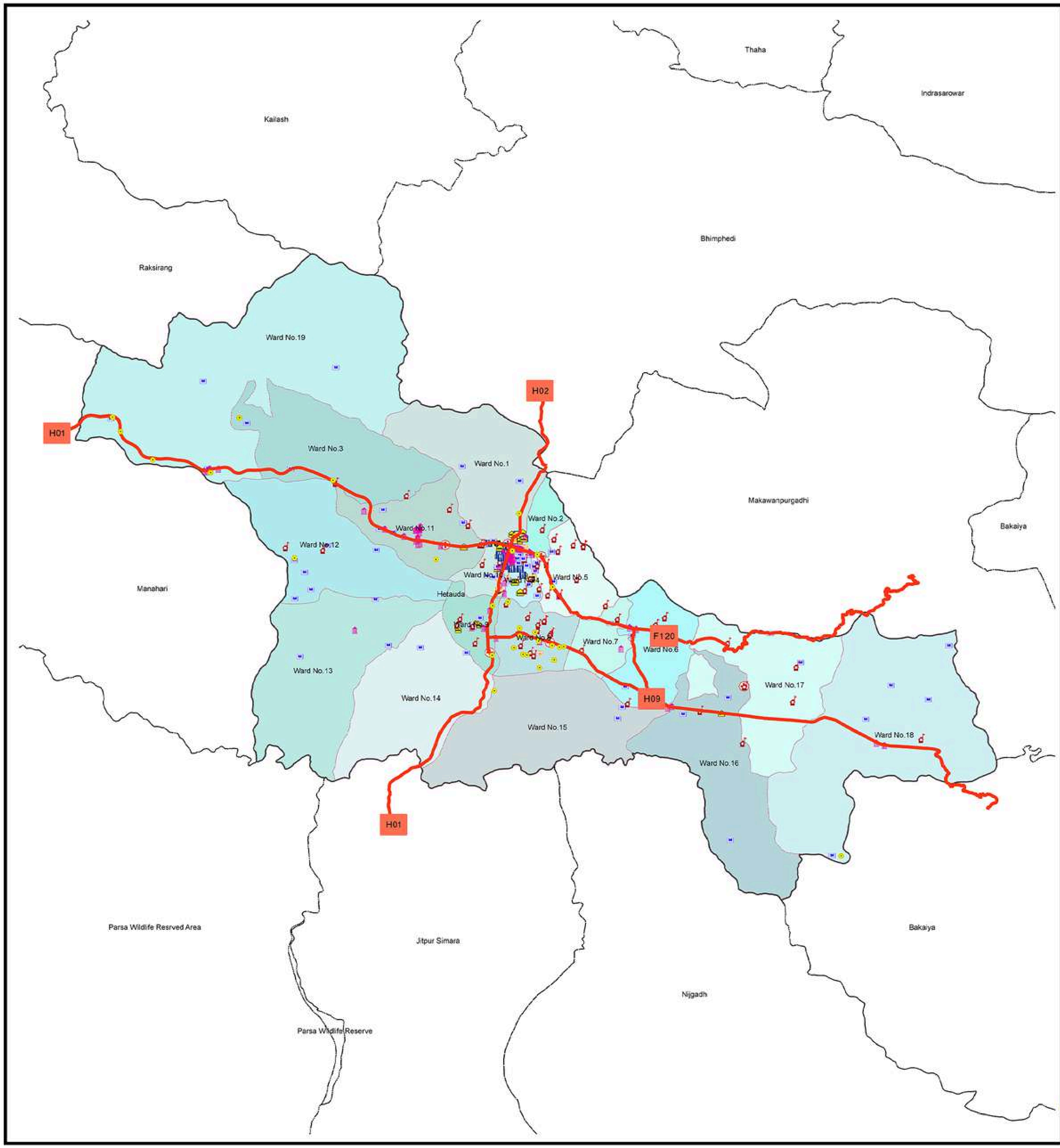


INTRODUCTION

Hetauda Sub-metropolitan City, located at coordinates 27°25′49′N , 85°01′56′E, serves as the administrative headquarters of Makwanpur District in Bagmati Province, Central Nepal. Declared the capital of Bagmati Province on January 12, 2020, Hetauda has grown into an important administrative and economic hub for the region.

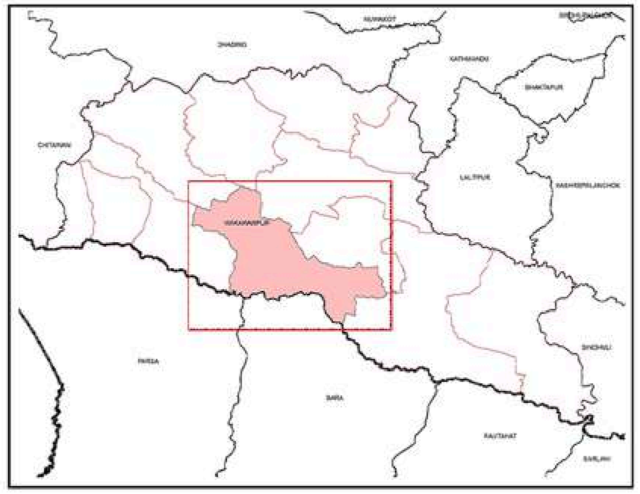
Nestled in a valley, Hetauda is surrounded by the Chure Hills to the south and the Midhills to the north, with the Mahabharat Range providing a lush and hilly landscape. Situated at an elevation of approximately 345 meters (1,132 feet) above sea level, the city is blessed with fertile plains, making it an ideal location for agriculture. The Rapti River, Karra River, and Kukhreni Khola flow through the area, contributing to the region's agricultural productivity and sustaining the livelihoods of its people. The climate is subtropical, with distinct wet and dry seasons, further supporting local farming activities.

Hetauda is also known for its industrial significance. The Hetauda Industrial District (HID), established in 1963, is one of the largest and most important industrial hubs in Nepal, hosting a wide range of industries from large-scale to cottage industries. The city's connectivity is facilitated by major roadways, including the East-West Mahendra Highway and Madan Bhandari Highway, which link Hetauda to other key areas in the country.



ADMINISTRATIVE MAP

Hetauda Sub-Metropolitan City, Makwanpur, Bagmati Province, Nepal



0 0.5 1 2 Kilometers

Coordinate System

Coordinate System: GCS WGS 1984
Datum: WGS 1984
Units: Degree

Data Source

1. Survey Department
2. Open Street Map (OSM)

Legend

- Provincial Boundary
- District Boundary
- Municipal Boundary
- Ward Boundary
- Settlement Area
- Village Area
- Highway
- Police Station
- Health Facility
- Educational Institute
- Financial Institute
- Bus Station
- Hotel and Restaurants
- Religious Site

Infograph

193,576 Total Population
207 KM² Population Density
261.58 KM² Total Area
95,678 Total Male
97,898 Total Female
46,566 Total Household

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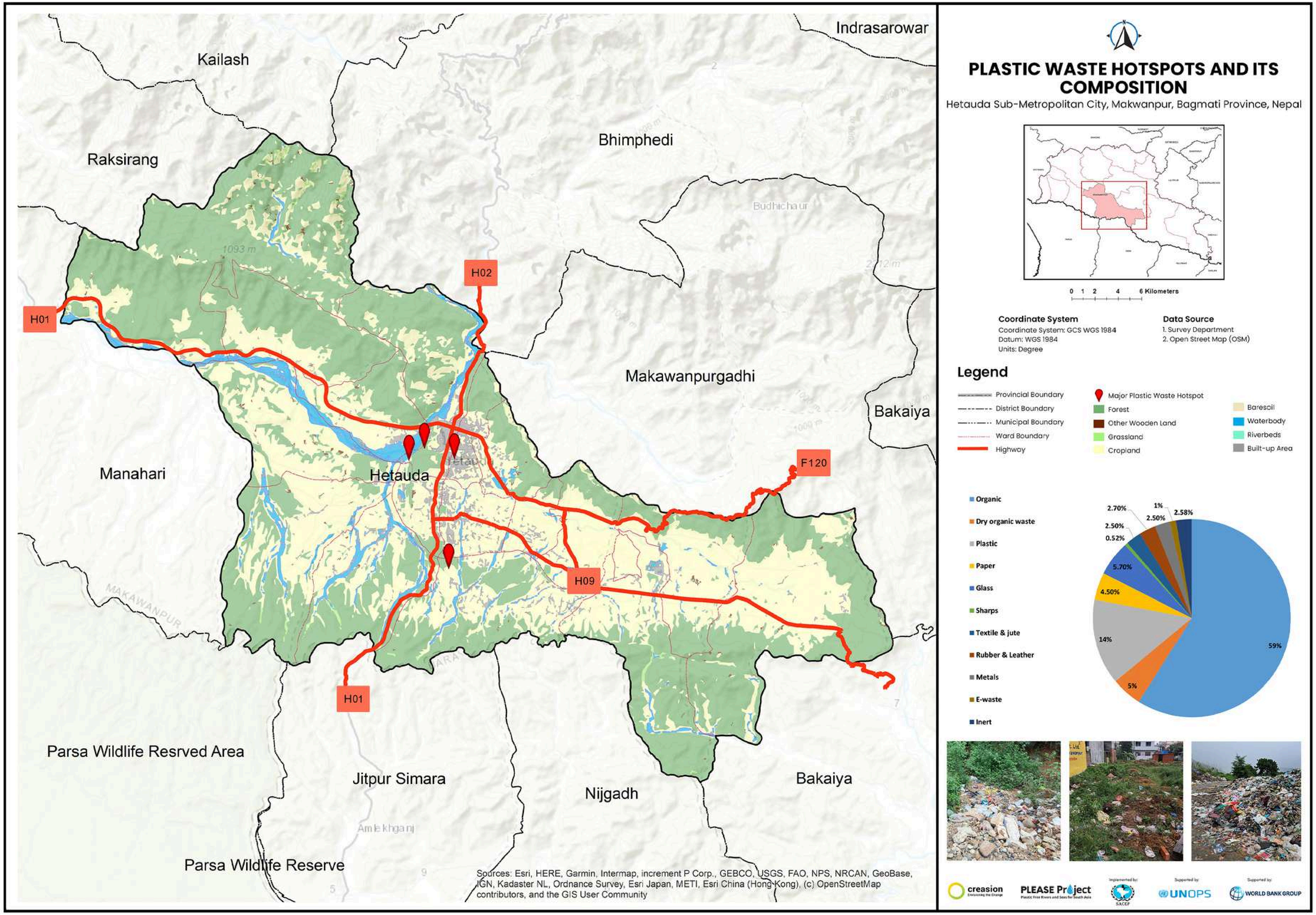
PLASTIC WASTE HOTSPOTS AND ITS COMPOSITION
Hetauda Sub-Metropolitan City, Makwanpur, Bagmati Province, Nepal
Kailash
Indrasarowar
Raksirang
Bhimphedi
Budhichaur
Makawanpurgadhi
Bakaiya
Manahari
Hetauda
H01
H02
H09
F120
MAKAWANPUR
Parsa Wildlife Resrved Area
Jitpur Simara
Nijgadh
Bakaiya
Amlekhganj
Parsa Wildlife Reserve
Sources: Esri, HERE, Garmin, Intermap, increment P Corp., GEBCO, USGS, FAO, NPS, NRCAN, GeoBase, IGN, Kadaster NL, Ordnance Survey, Esri Japan, METI, Esri China (Hong Kong), (c) OpenStreetMap contributors, and the GIS User Community
0 1 2 4 6 Kilometers
Coordinate System
Coordinate System: GCS WGS 1984
Datum: WGS 1984
Units: Degree
Data Source
1. Survey Department
2. Open Street Map (OSM)
Legend
Provincial Boundary
District Boundary
Municipal Boundary
Ward Boundary
Highway
Major Plastic Waste Hotspot
Forest
Other Wooden Land
Grassland
Cropland
Baresoil
Waterbody
Riverbeds
Built-up Area
Organic
Dry organic waste
Plastic
Paper
Glass
Sharps
Textile & jute
Rubber & Leather
Metals
E-waste
Inert
59%
5%
14%
4.50%
5.70%
0.52%
2.50%
2.70%
1%
2.50%
2.58%
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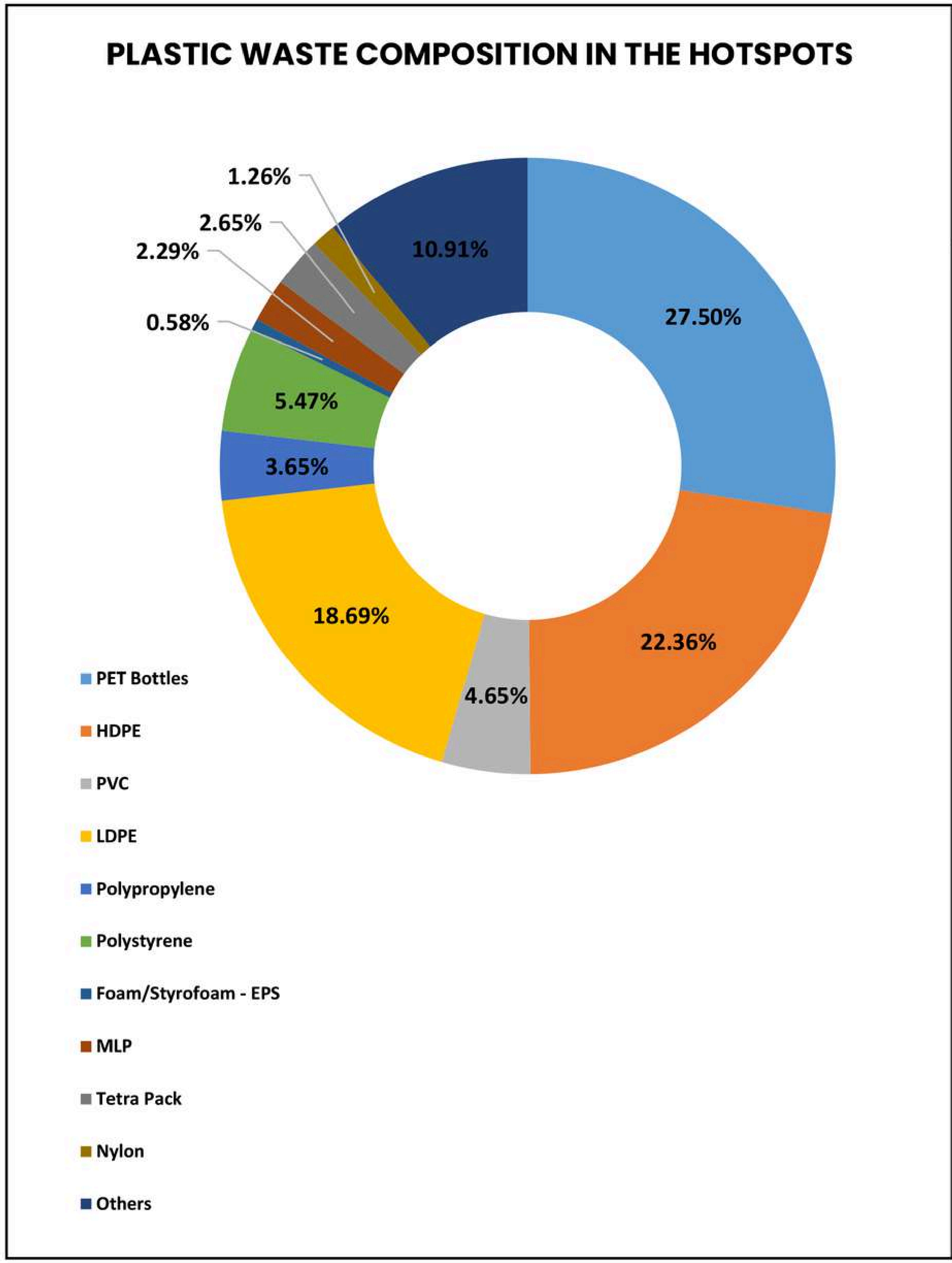
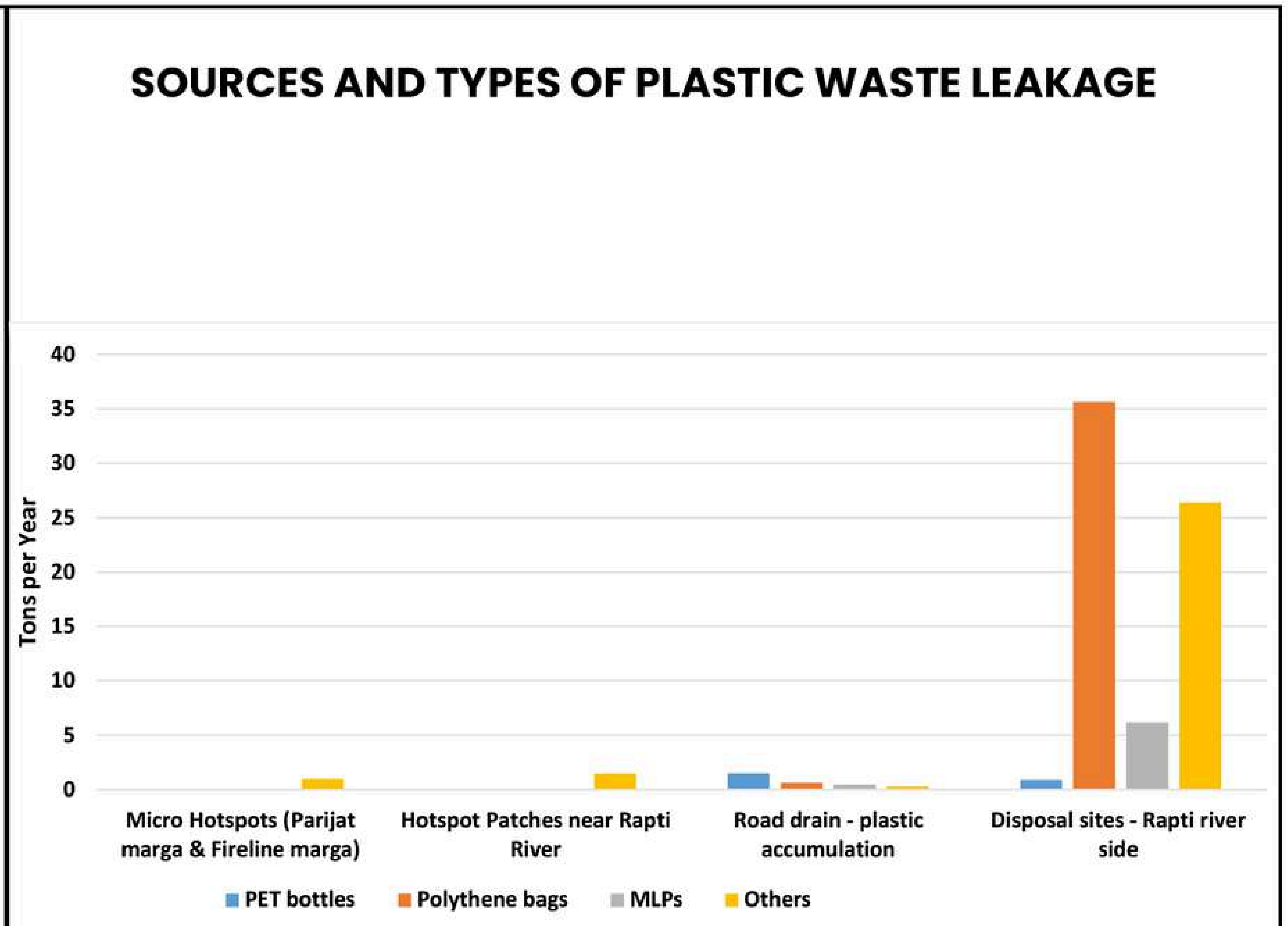
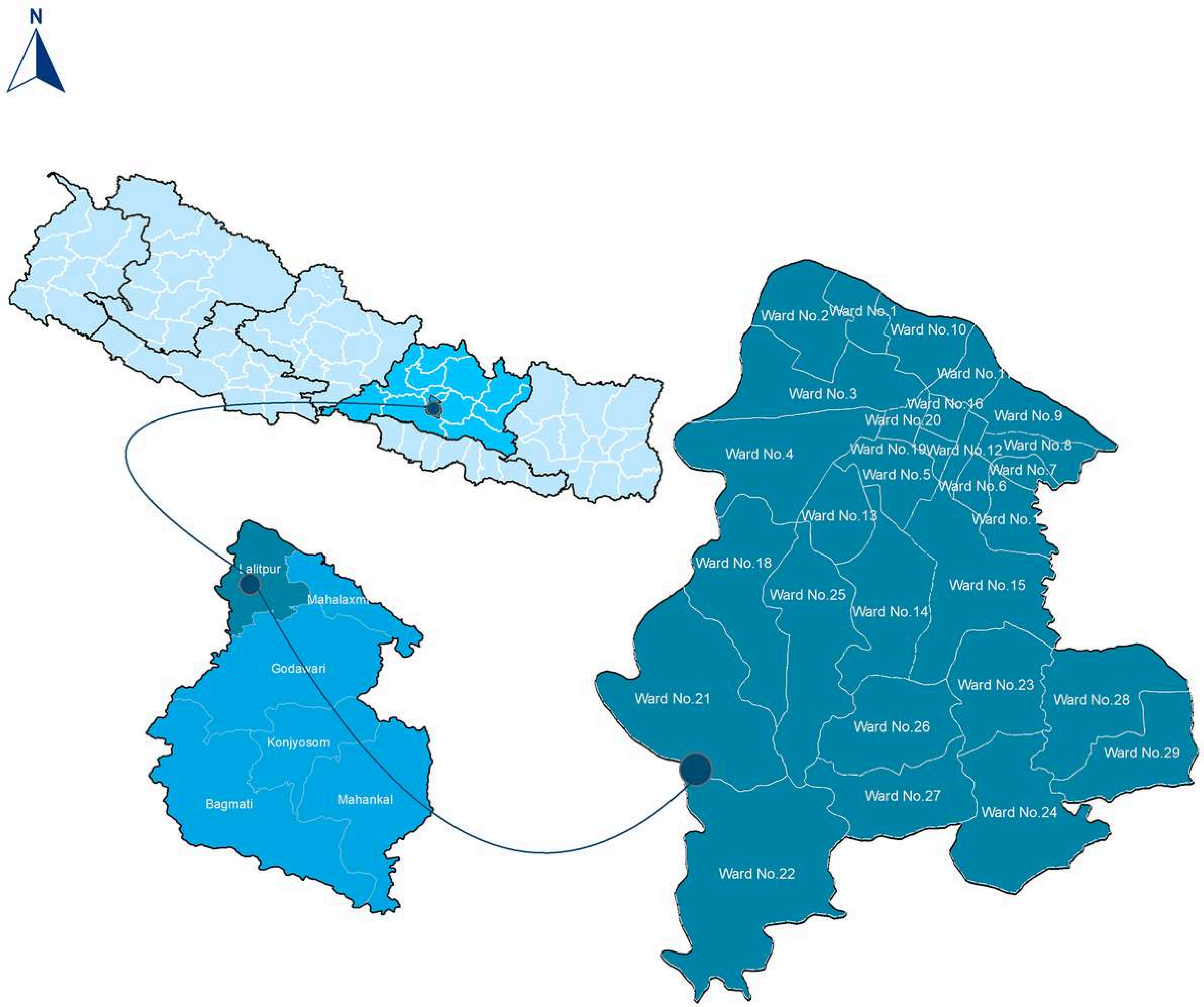

PLASTIC WASTE COMPOSITION IN THE HOTSPOTS
1.26%
2.65%
2.29%
0.58%
10.91%
27.50%
5.47%
3.65%
18.69%
22.36%
4.65%
PET Bottles
HDPE
PVC
LDPE
Polypropylene
Polystyrene
Foam/Styrofoam - EPS
MLP
Tetra Pack
Nylon
Others


SOURCES AND TYPES OF PLASTIC WASTE LEAKAGE
Tons per Year
40
35
30
25
20
15
10
5
0
Micro Hotspots (Parijat marga & Fireline marga)
Hotspot Patches near Rapti River
Road drain - plastic accumulation
Disposal sites - Rapti river side
PET bottles
Polythene bags
MLPs
Others

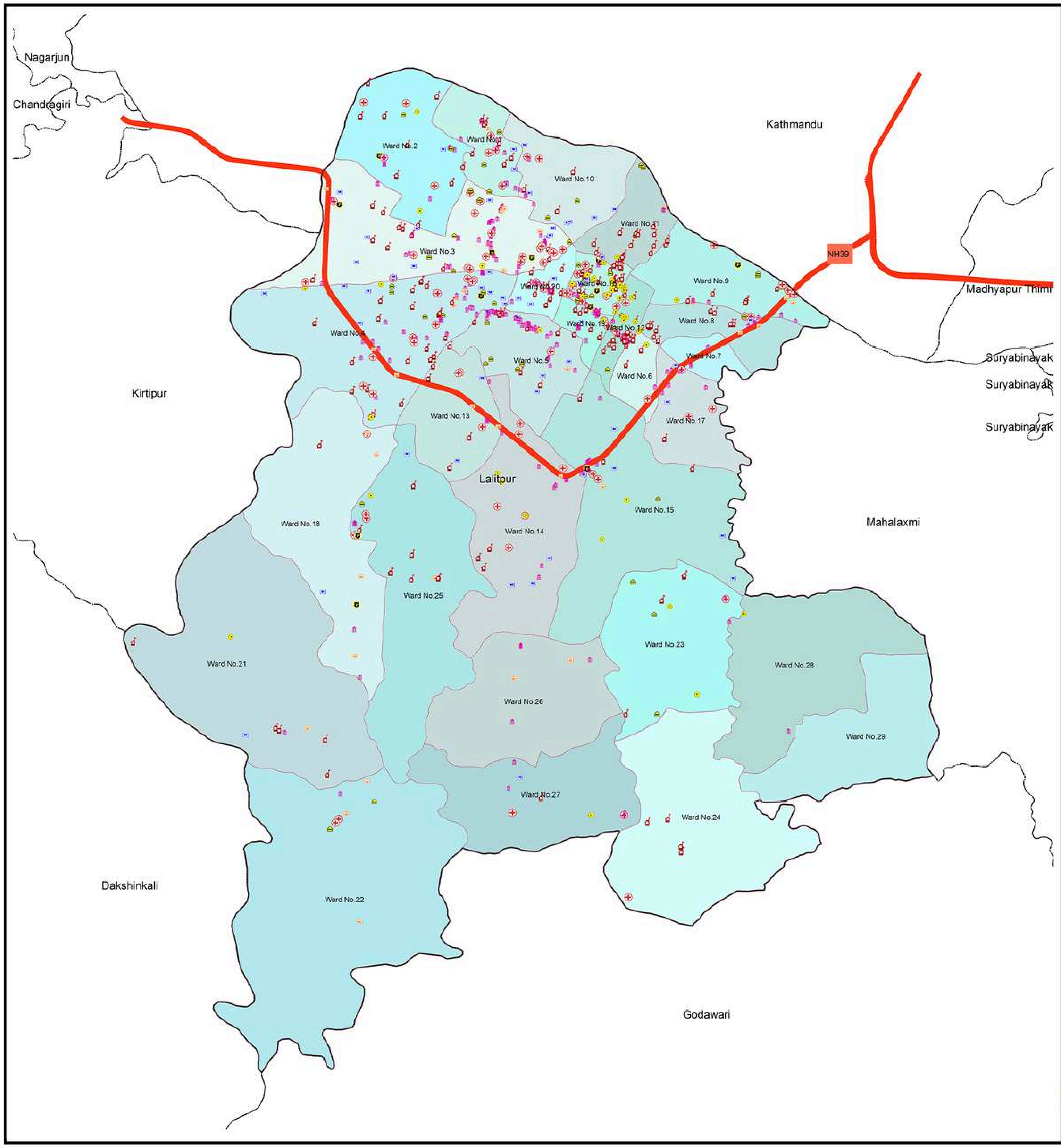


INTRODUCTION

Lalitpur Metropolitan City, one of the major urban centers in Nepal, is located in the heart of the Kathmandu Valley, renowned for its rich history, vibrant culture, and stunning architecture. According to the 2021 census, Lalitpur has a population of 294,098, with 49,044 households and a literacy rate of 90%. Covering an area of 395.08 square kilometers, the city is divided into 29 wards and sits at an altitude of approximately 1,400

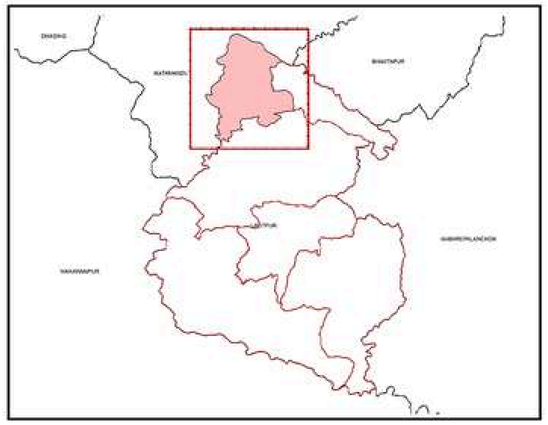
Lalitpur is a thriving hub for education and culture, known for its harmonious blend of Hindu and Buddhist communities. It is famous for its festivals, religious events, and historical landmarks, including the UNESCO World Heritage Site, Patan Durbar Square. The city's mild climate, with average temperatures ranging from 12°C to 28°C, ensures comfort throughout the year.

A significant portion of the population is engaged in crafts, tourism, commerce, and agriculture, all of which contribute to the city's dynamic economy. Lalitpur is also home to several important rivers, including the Bagmati River, which holds deep religious and cultural significance. Along with Lalitpur Khola and Taukhel Khola, these rivers play a crucial role in the region's water drainage system and support the local ecosystems.



ADMINISTRATIVE MAP

Lalitpur Metropolitan City, Lalitpur, Bagmati Province, Nepal



0 0.5 1 2 Kilometers

Coordinate System

Coordinate System: GCS WGS 1984
Datum: WGS 1984
Units: Degree

Data Source

1. Survey Department
2. Open Street Map (OSM)

Legend

- Provincial Boundary
- District Boundary
- Municipal Boundary
- Ward Boundary
- Settlement Area
- Village Area
- Highway
- Police Station
- Health Facility
- Educational Institute
- Financial Institute
- Bus Station
- Hotel and Restaurants
- Religious Site

Infograph

294,098	Total Population
1433 KM²	Population Density
36.12 KM²	Total Area
146,730	Total Male
147,368	Total Female
77,159	Total Household

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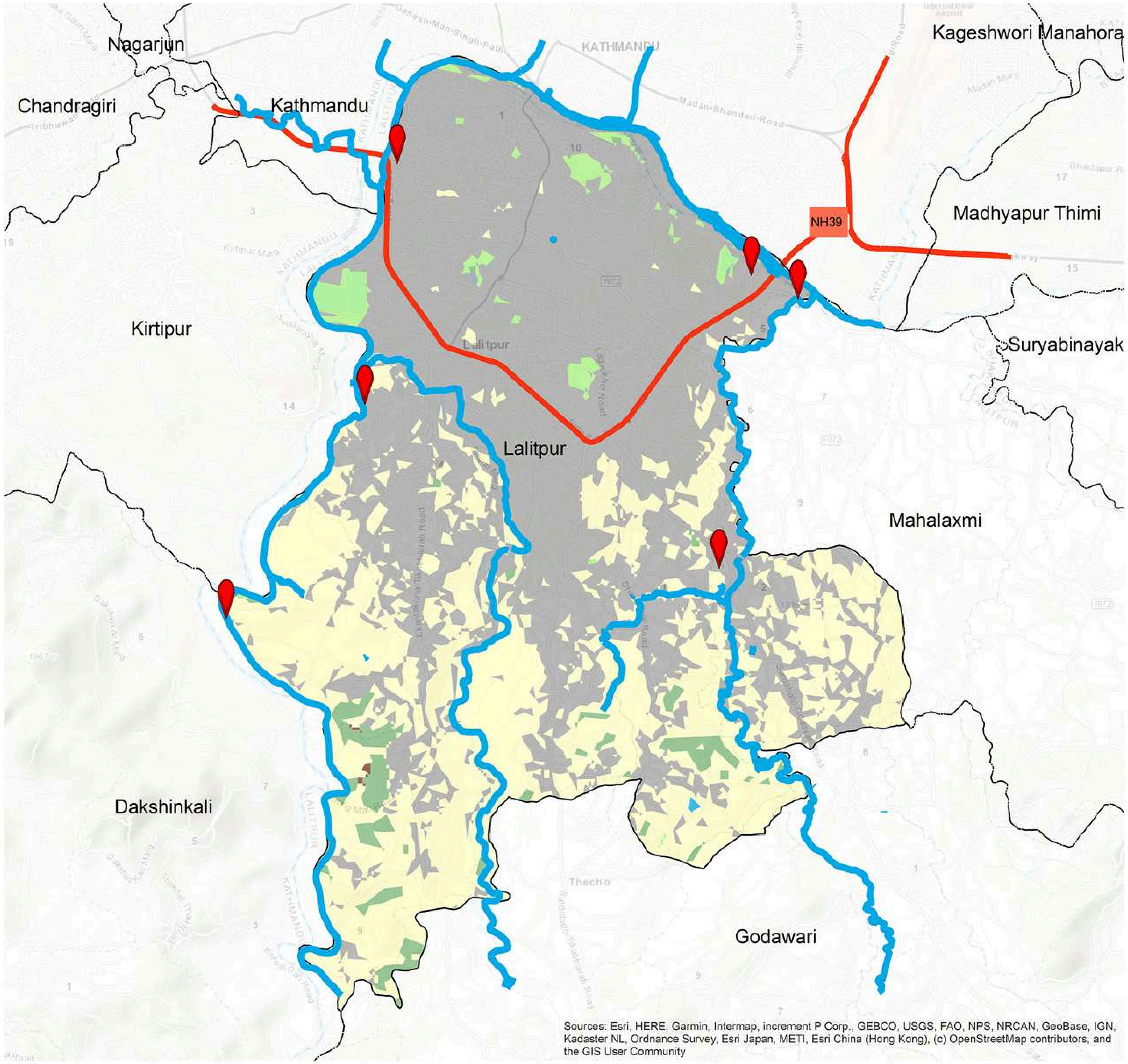
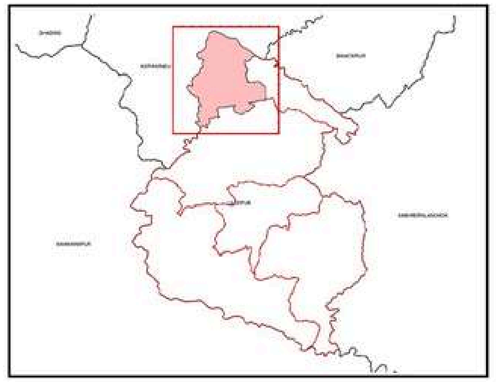
PLASTIC WASTE HOTSPOTS AND ITS COMPOSITION

Lalitpur Metropolitan City, Lalitpur, Bagmati Province, Nepal





0 1 2 4 6 Kilometers

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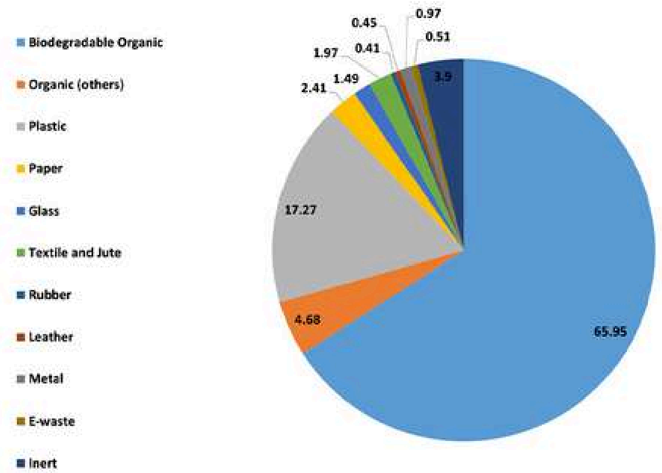

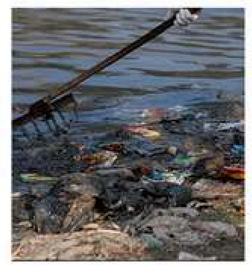

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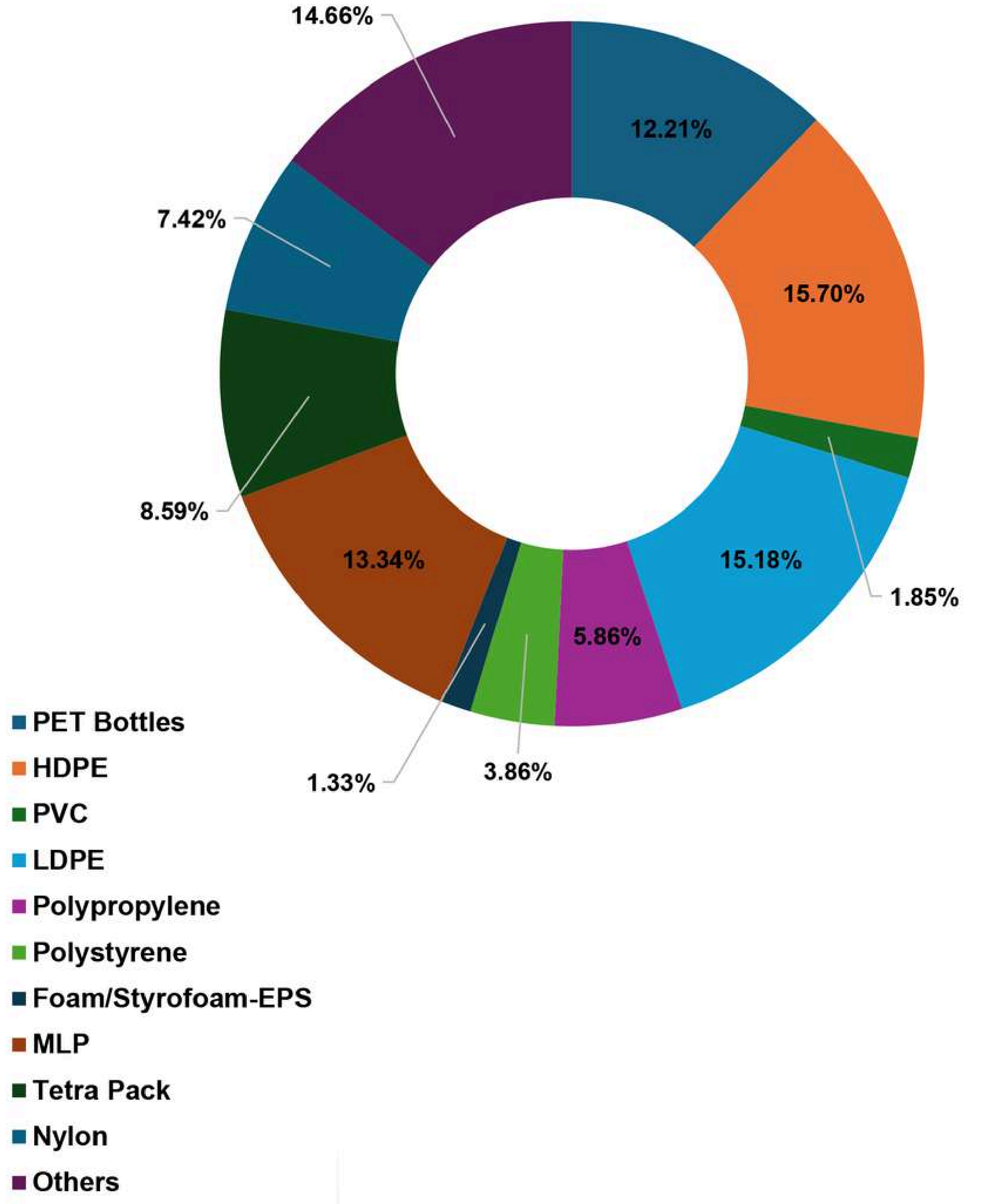
PLASTIC WASTE COMPOSITION IN THE HOTSPOTS



SOURCES AND TYPES OF PLASTIC WASTE LEAKAGE